



DaVinci 200

User Manual

• DOCUMENT MANAGEMENT

Document History

Version	Status	Date	Responsible	Reason for Change
1.0	Release	31.03.2022	NS	Initial Release
1.1	Release	05.04.2022	DW	Changes in: Introduction of DaVinci Loadport Monitoring Verschieden Themen
1.2	Release	05.04.2022	DW	Correction images and text
1.3	Release	05.04.2022	DW	Topics: Prealigner, Interlock, Loadports, Gem Status
1.4	Release	06.04.2022	DW	Recipe in Operator Modus and supervisor
1.5	Release	07.04.2022	DW	Recipe copy with TC und Nanostar
1.6	Release	07.04.2022	DW	Rezept copying and reteaching 5.
1.7	Release	08.01.2025	DW / SAz	Changes for DaVinci 200
1.8	Release	10.01.2025	SAz	Updated images, proof-reading

Legal Notices and Addresses

© 2025 MueTec Automatisierte Mikroskopie und Messtechnik GmbH
All rights reserved.

Company Headquarters

MueTec GmbH
Hans-Bunte-Str. 5
D-80992 München
Germany

Telephone: +49 (0) 89/1500160-0
Fax: +49 (0) 89/1500169-60

E-mail: sales@muetec.com
Web: www.muetec.com

Production Plant

MueTec GmbH
Isarauer Str. 77
D-94527 Aholming
Germany

Telephone: +49 (0) 9938/9191-0
Fax: +49 (0) 9938/9191-13

E-mail: sales@muetec.com
Web: www.muetec.com

Sales and Application Support for Asia

MueTec Taiwan Ltd.
No. 229, 8F – 7. Fu-Xing 2nd Road
JhuBei City
Hsinchu County
Taiwan 30271
R.O.C.

Telephone: +886 972 718 699
Fax: +886 355 010 92

Publication ID:

DaVinci200-UM-WEP-250110

Content

•	DOCUMENT MANAGEMENT	2
1	Introduction	6
2	Safety Instructions	7
2.1	Personal Requirements	7
2.2	Warning Symbols	7
2.3	General Hazards and Precautions	8
2.3.1	General Safety Instructions	8
2.3.2	Mechanical Hazards	9
2.3.3	Electrical Hazards	9
2.4	Special Precautions	10
2.4.1	Transport	10
2.4.2	Installation	10
2.4.3	Commissioning	11
2.4.4	Industrial Safety	11
2.4.5	Operation	12
2.4.6	Life Cycle	12
2.4.7	Disposal	12
2.5	Malfunctions and Emergencies	13
3	System Description	14
3.1	Overview	14
3.2	Signals and Controls	16
3.3	Online Visualization / Touchscreen / User Screen	17
3.3.1	Overview	17
3.3.2	Online Visualization: Module Selection and Online Button	18
3.3.3	Admin GUI: Supervisor and/or Service Permissions	21
3.3.4	Visualization: System Overview	22
3.3.5	Online Visualization: Operator GUI	26
3.3.6	Screen Area Operator GUI: GEM Status, System Overview, LP Overview	28
3.3.7	Substrate Information	32
4	Operation	33

4.1	Overview	33
4.2	Modes of Operation	33
4.3	System Messages	34
4.4	Starting the Software and Initializing the System	34
4.5	Setting Up Cassettes and Starting a Job	38
4.6	Loading FOUP/Cassette/SMIF and Setting CarrierID in GEM Mode Offline and Local Online	40
4.7	Job Creator and Start of a Measurement Cycle	42
4.8	Interrupting Jobs	49
5	Recipes	50
5.1	Login as Supervisor	50
5.2	Starting the Recipe Manager	51
5.3	Managing Recipes	52
5.3.1	Edit Recipe	52
5.3.2	Copy Recipe or Create New Recipe	52
5.3.3	Rename Recipe	52
5.3.4	Delete Recipe	52
5.4	Creating or Changing Recipes (Detailed)	53
5.4.1	Create a new Folder	53
5.4.2	Delete a Folder	54
5.4.3	Copy a Recipe	55
5.4.4	Edit a Recipe and Re-teaching	58
5.4.5	Delete a Recipe	61
5.4.6	Rename a Recipe	61

1 Introduction

This user manual explains the applications of the DaVinci 200 system.

DaVinci 200 is an automated high end metrology tool with measurement options for critical dimensions (CD optional), overlay (OVL optional) and film thickness (FT optional) measurements of 200 mm (150mm optional) wafers. These instructions and the additional operating instructions, e.g. *ToolCommander User Manual*, provide all necessary information for safe and successful operation of the system.

About MueTec

MueTec develops high precision optical measurement and inspection solutions that help customers in the semiconductor and microelectronics industries to improve their manufacturing yield and overall process control. Our team of technical experts knows, fulfills and anticipates the needs of the global producers of wafers, masks and MEMS and offers custom-tailored and standard solutions that distinguish themselves through their extremely low maintenance requirements. Scalable systems allow for a combination of many different applications in a single system, including critical dimension, overlay and film thickness measurement as well as defect inspection and review.

In the last 25 years, MueTec has installed 300 metrology and inspection systems worldwide. There is hardly a leading semiconductor manufacturer that does not use our technology in production. Due to our ability to adapt systems precisely to customers' needs, our customers include not just large semiconductor companies, but also research institutions all around the world.

2 Safety Instructions

This chapter contains important information on safety aspects. Read and understand this chapter **before** starting to operate the system. For your own safety, you should also read and understand all other instructions before commissioning or booting the system.

2.1 Personal Requirements

Operators, supervisors and service personnel of the system must be instructed in the operation and maintenance of the system by MueTec training personnel or by other competent and authorized persons before carrying out any work on or with the system themselves.

In addition, a thorough knowledge of wafer handling and safe behavior in a production environment and cleanroom is a prerequisite for using the system.

Only authorized and instructed personnel may operate the system. Do not remove any parts of the protective edge of the system! Contact MueTec before making any changes to the device or installing any unauthorized components or components not specified by MueTec. System maintenance and repairs may only be carried out by MueTec's technical service or by personnel instructed and authorized by MueTec. Only authorized personnel (electricians) may open the electronics housing and the housing cover.

2.2 Warning Symbols

The notes and warnings attached to the system must be carefully observed.

This sticker indicates the danger of moving parts:



The following stickers indicate that the mounted laser is based on laser class 1 and laser class 3B (in case of coarse opening of the laser module), which entails the necessary safety measures:



2.3 General Hazards and Precautions

To ensure safe maintenance, please read and follow the following instructions carefully.

2.3.1 General Safety Instructions



Never reach into the system. The tasks of an operator must be performed from outside the system (e.g. setting up a FOUP/cassette/SMIF, starting jobs). If an intervention inside the system appears necessary, inform your supervisor and/or the responsible service personnel.



Never touch moveable or moving parts of the system.



Do not touch wafers by hand under any circumstances.



Avoid shaking or moving the system.



The housing doors on the right side of the system may only be opened when the system is switched off (main switch off). This is reserved for service personnel only.



System operators must not open any doors on the system and must not remove any parts of the fairing. By opening a door or by opening cladding parts, an emergency stop occurs, which is why you need a supervisor or a service technician for the further process sequence.



Power and vacuum connections must not be made, manipulated or removed by the system operator.



Cables and lines in the system must not be connected, manipulated or removed by the operator.



Operate the touchscreen only with your fingers or an approved and suitable object.



Use a mouse and keyboard for operation.



System error messages must be carefully read, understood and processed. If you have any questions, please contact your supervisor.



Do not reach between the load port and the FOUP/cassette/SMIF.



If the system informs you that a double slot allocation or a cross slot is detected in the FOUP/cassette/SMIF, check this. Do not continue working but correct the wrong placement by yourself or a qualified person.



No transport trays may be used on the system. Transport trays can lead to the destruction of robot parts or the load port. In addition, damage to wafers cannot be ruled out.

2.3.2 Mechanical Hazards



Risk of injury from moving parts. Do not reach into the system or parts of the system if parts are moving.

2.3.3 Electrical Hazards



Danger from live parts

The system includes live components, cables and connectors. Do not touch such parts with your hand or with unsuitable objects.

The control cabinet doors may only be opened by qualified persons with suitable training and appropriate education. Persons must be able to identify risks and avoid hazards that may emanate from electricity (electricians). See also EN 60204-1 – electrical equipment of machines – part 1, 3.53.

For electrical work on the system, the specifications of BGV A3 "Unfallverhütungsvorschrift für elektrische Anlagen und Betriebsmittel" must be observed. Inspection intervals according to §5 of the BGV A3 must be specified for the system.

2.4 Special Precautions

The following instructions and measures apply to the entire life cycle of the system from transport, commissioning and operation to disposal.

The instructions must be noted and strictly observed by the operator of the system as well as by all people who come into contact with the system (supervisor, operator, maintenance and service personnel). Careful observance of the instructions is essential for proper and safe handling of the system.

2.4.1 Transport

- The machine must be disconnected from the energy sources connected to the machine before any transport movement. It must be ensured that no additional electrical or fluidic hazards can arise as a result of energy separation. The energy separation must not take place under load, but only without voltage and pressure.
- Assembly, disassembly, commissioning, decommissioning and disassembly must be carried out by specially trained personnel. The operating personnel must be informed about all residual risks related to assembly and disassembly. The staff shall meet the following minimum requirements:
 - Knowledge and, if necessary, approval and qualification for proper load transportation (e.g., crane operation, operating authorization for cranes)
 - Knowledge of the load distribution of the entire machine or the load to be transported
 If the specialist knowledge is not available, it is advisable to commission a specialist company for the transport. The responsibility for this lies with the operator of the machine.
- For machine transport by separate load carriers (means of transport), the following applies: the machine and associated components may only be transported using suitable vehicles, cranes, forklifts or lift trucks (depending on the dimensions and weight of the system components). The means of transport to be used must be able to withstand the expected stress and load and must be sufficiently dimensioned. The transport routes must be sufficiently dimensioned to exclude possible collisions between the transported load and the transport environment (hall boundaries, passage openings, other machines, etc.).
- The lifting equipment used (cranes, ropes, slings, etc.) must be sufficiently dimensioned for the total loads to be moved and able to withstand the expected loads. Loads must always be lifted horizontally. It must be ensured that the load is always in equilibrium. The machine may only be attached to the attachment points marked by the manufacturer.
- Machines and machine components to be moved may only be lifted as high as is necessary for transport. An unnecessarily high transport of the load increases the risk of falling and possible machine damage.

2.4.2 Installation

- The installation and assembly of the machine and associated machine components may only be carried out by personnel named and instructed by the machine manufacturer.
- In direct consultation with the machine manufacturer, the persons are informed of possible risks and carry out assigned activities only according to instructions. It is mandatory to observe
 - that the personnel are appropriately trained for installation and assembly
 - that breaks and rest periods are taken into account
 - that for the activities necessary, appropriate and approved tools are used
 - that (if necessary) personal protective equipment is used in accordance with the instructions of the operator or the present workplace conditions or from the present activity.
- The personnel carrying out installation and assembly work must obtain information from the machine operator about risks arising from the working environment or the planned location of the machine.
- The focus module must not be switched on when disassembled. There is a class 3B laser (infrared laser) in the focus module. The direct view into the laser can damage the eyes.
- The focus module may only be removed when the main switch is switched off.

2.4.3 Commissioning

- Commissioning of the machine functions, associated settings (adjustments, parameters, programming) as well as test runs of the machine and individual machine components required in connection with commissioning may only be carried out by qualified personnel who have been designated by the machine manufacturer and who have been instructed accordingly.
- In direct consultation with the machine manufacturer, the persons are informed of possible risks and carry out assigned activities only according to instructions. It is essential to ensure
 - that the personnel for installation and assembly are appropriately trained
 - that breaks and rest periods are taken into account
 - that for the activities necessary, appropriate and approved tools are used
 - that (if necessary) personal protective equipment is used in accordance with the instructions of the operator or the present workplace conditions or from the present activity.
- The personnel carrying out installation and assembly work must obtain information from the machine operator about risks arising from the working environment or the planned location of the machine.
- The focus module must not be disassembled nor opened.
- The focus module must not be switched on when disassembled. There is a class 3B laser (infrared laser) in the focus module. The direct view into the laser can damage the eyes.
- The focus module may only be removed when the main switch is switched off.

2.4.4 Industrial Safety

- The operator must ensure that all people constantly working with the machine take sufficient breaks to recover and that there is sufficient fresh air supply to prevent signs of fatigue and lack of concentration when working with the machine.
- The operator must ensure that the working areas of the machine and the direct machine environment remain free of obstacles, which could endanger the machine personnel. Objects and materials required for production or maintenance work must either be placed in the designated positions or placed in such a way that no immediate hazards can arise. If necessary, unavoidable hazardous conditions that can occur in the machine environment must be identified by suitable measures (e.g. fencing off with red and white barrier tape, information signs, etc.).
- Escape routes must in any case be marked and kept free of objects that could impede escape.

2.4.5 Operation

- Before starting any activity in the working environment of the machinery, the staff designated for that purpose must ensure that the floor conditions meet the requirements and that the working environment is safe. Among other things, it must always be ensured
 - that leaked liquids, aerosols or other impurities which may cause a slipping hazard, are removed before carrying out an activity in this area
 - that the working area is sufficiently dimensioned to allow the work to be carried out without tripping or impacting edges in the working area
 - that unavoidable impact and tripping points are secured or covered and conspicuously marked.
- The surrounding area of the machine, in particular the working areas, must be cleaned at regular intervals. All necessary measures for the protection of people must be taken by the operator (e.g. information signs for danger of slipping in case of damp floor cleaning).
- The physical condition and the control effect of the emergency-off devices on the entire machine must be checked at regular intervals (recommendation: at least once a day) in order to ensure the constant availability required by the Machinery Directive. The inspections carried out must be documented by the operator. If the emergency stop function fails, the machine must be stopped immediately. The machine may only be restarted if the emergency stop function is guaranteed again.
- The machine has a monitoring of the housing. The cassette monitoring (Interlock) means that it is not possible to control the robot as soon as a cassette of the outer housing is removed. With an interrupted door safety circuit, the robot can only be operated with the teach panel.

2.4.6 Life Cycle

- The maximum service life of the system is 10 years starting from acceptance. If the service life of individual parts and components (e.g. safety-relevant components) is limited by the part or component manufacturer, this limitation shall also apply to the entire system.

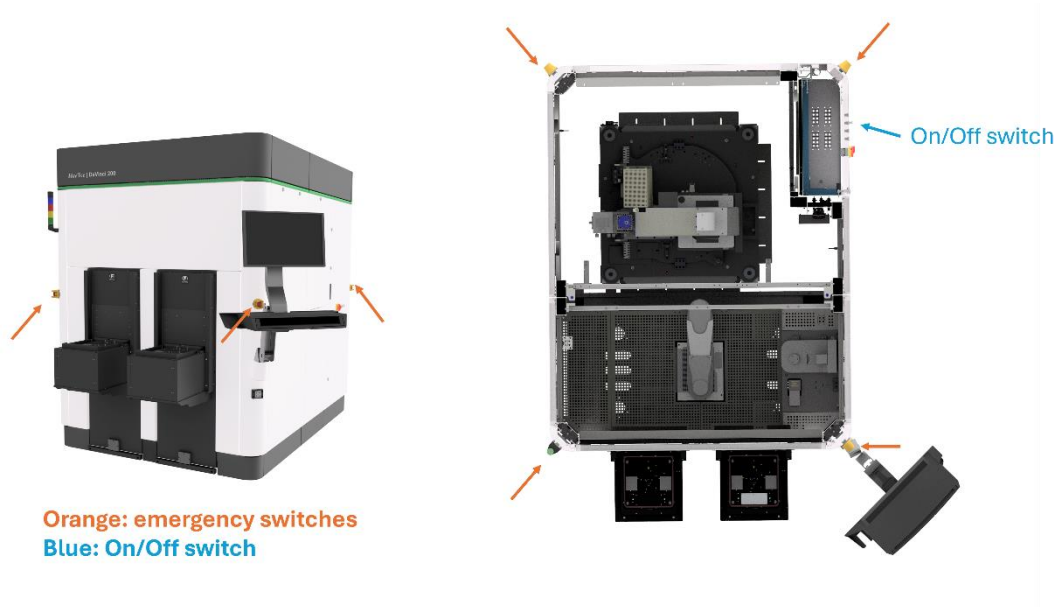
2.4.7 Disposal

- For the decommissioning of the machine, it is imperative to observe in the first step,
 - that the machine is to be switched off via the main disconnection device
 - that all power supplies (electricity, compressed air, etc.) are interrupted by qualified personnel and separated from the machine
 - that access to the machine is restricted or prevented.
- The machine and the associated components must be dismantled by a suitable specialist company. For this purpose, the information from the life cycle phases of assembly and commissioning must be observed and implemented.
- The machine and the associated components must be disposed of by a suitable specialist company. The national regulations applicable at the time of disposal at the location of the machine and its components must always be observed.

2.5 Malfunctions and Emergencies

The system is equipped with four emergency switches, which immediately interrupt all movements of the robot/transport module.

Position of the emergency switches and the main on/off switch:



In case of an emergency

Press the emergency stop button when

- there is a danger to persons working on or near the system
- a mechanical failure could damage the system or the wafers.

Inform the supervisor responsible immediately. The system may only be checked and re-started by a supervisor.

Incorrectly placed wafers

- If you detect an incorrectly placed wafer in a cassette (e.g. in a tilted position), correct its incorrect position immediately before continuing the inspection process.
- If the system detects incorrectly placed wafers, remove the cassette, correct the position, and reset the cassette. If you are unsure about the correct wafer position, ask the supervisor.

In case of malfunctions



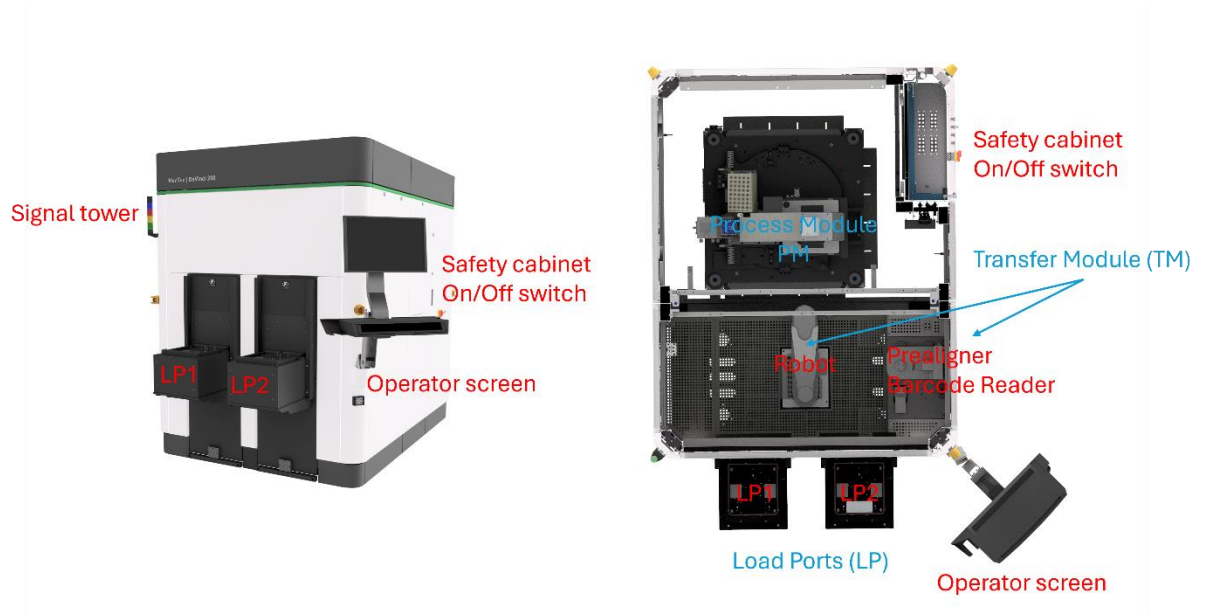
Opening all doors and system paneling immediately shuts down the system. The main switch must be switched off for maintenance and electrical work on the system. Maintenance work and work on electrical components may only be carried out by trained and authorized personnel.

3 System Description

This chapter gives an overview of the main components of the system. In particular, the signal and operating elements are described here. In addition, assistance for system problems is shown.

3.1 Overview

The following illustration shows all system components relevant to the operator.

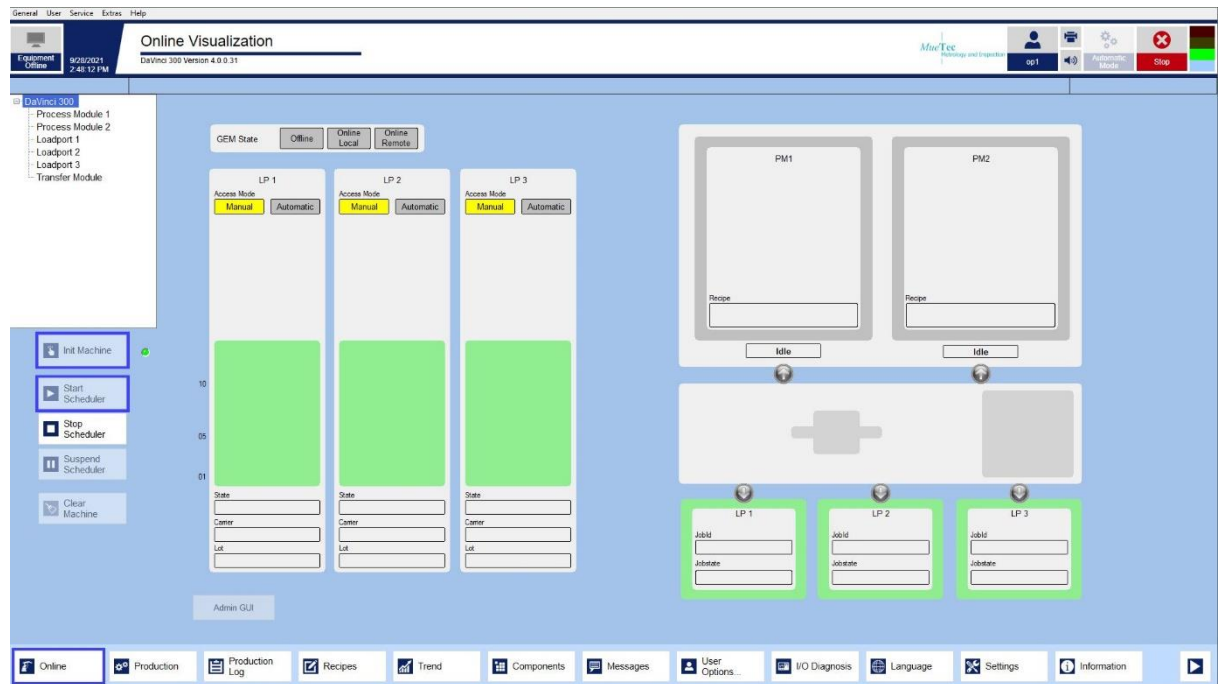


LP = Loadport, TM = Robot and Preamplifier, PM = Process Module

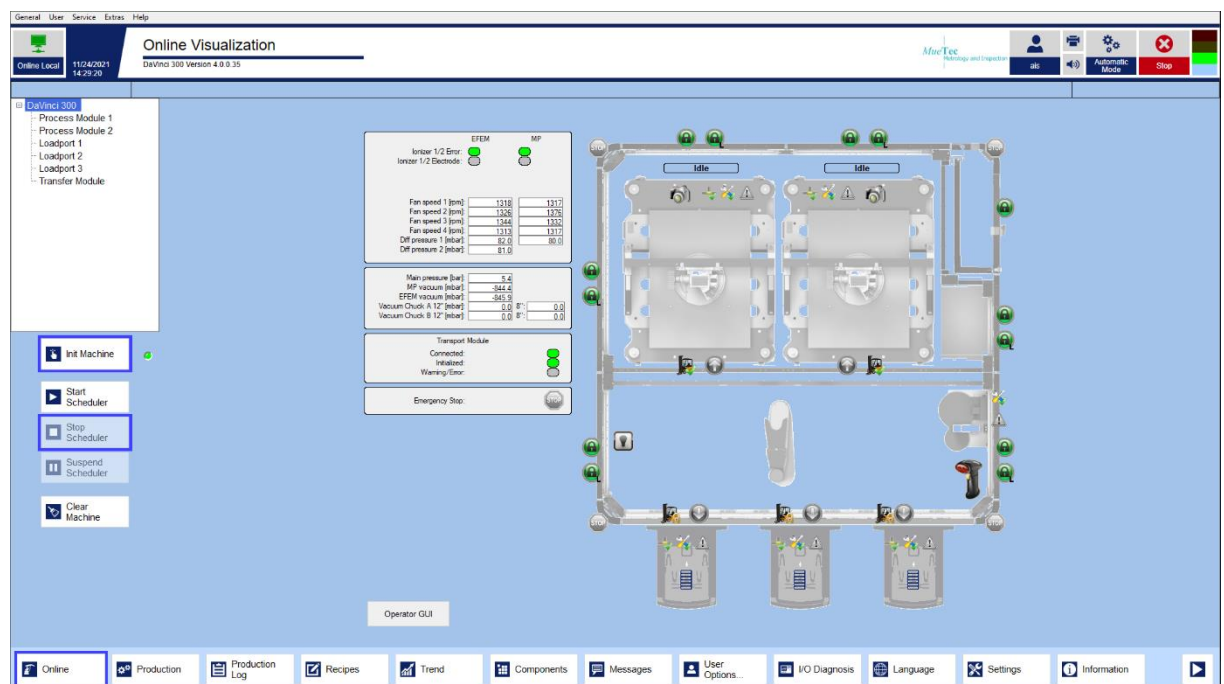
The system can be equipped with 1, 2 or 3 LPs and with 1 or 2 PMs (optional, dependent on stage and customer).

The touchscreen as user interface can be mounted left or right of the load ports. Due to the technical conditions of the working environment, it is recommended to enter data with a mouse and keyboard. In the online visualization of the user screen, the modules are displayed in such a way that operation is as simple and intuitive as possible.

Operator GUI with operator permissions:



Admin GUI with supervisor and/or service permissions:



3.2 Signals and Controls

The signal light

The signal light has acoustic and visual signal elements. The table shows the standard configuration of the signal elements and their meaning. The signal elements can be customized. If you are not sure, contact your supervisor.



Signal	Meaning
Acoustic	Malfunction! Ask the supervisor for help!
Blue light	The system runs in automatic mode (SECS/GEM protocol).
Red Light	Malfunction! Ask the supervisor for help!
Yellow light	Warning! Note the messages displayed! If you have any questions, contact your supervisor.
Green light	Normal operation. No malfunctions.

Main switch

The main switch turns the system on and off.



The left button lights up when the system is on. The system should only be switched off and on by the supervisor or an instructed person.

Emergency stop button

The emergency stop switches off the power supply to the load ports, the stage and the robot in emergencies. See also chapter 2.5 Malfunctions and Emergencies.

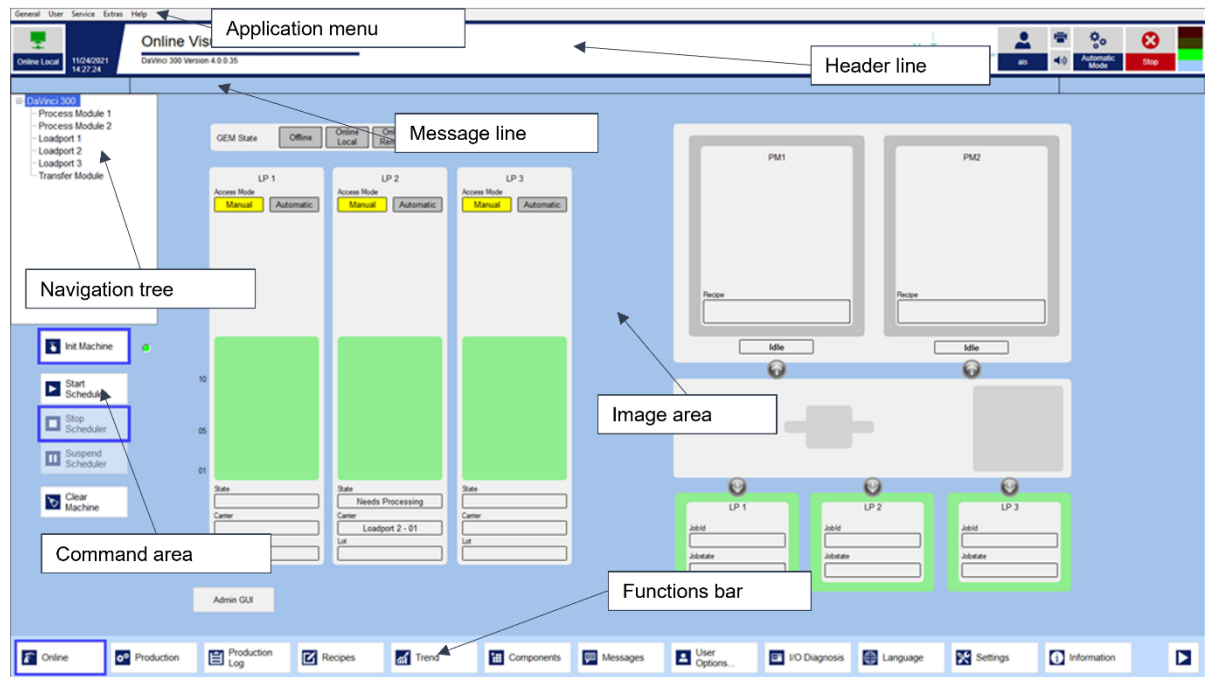


3.3 Online Visualization / Touchscreen / User Screen

3.3.1 Overview

All processes of the system are controlled by software (ToolCommander®). For safe handling and detailed knowledge, please also read the *Software Operation Manual* and *TC User Documentation*.

General screen layout:



The user interface is displayed on the touchscreen. To make operation easier, there are various user levels making several functions are inaccessible (buttons grayed out). You can only operate buttons that are not grayed out.

There are two different representations of the overview screen:

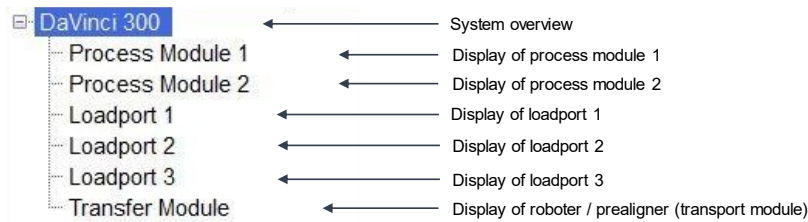
- The Admin GUI for the **Service** and **Supervisor** user levels
- The Operator GUI for the **Operator** user level

The GUIs differ mainly in the screen area and the view, but also in available functions in this area.

Operator, supervisor and service personnel use the system's touchscreen or mouse and keyboard to communicate with the software's user interface. Each type of user of the system has its own combination of ID and password to log in to the software. The touchscreen is mounted on a movable arm on the side of the system. The touchscreen is operated with the fingertips or another suitable instrument. Due to the conditions of the working environment (cleanroom) it is recommended to enter the data directly with mouse and keyboard.

3.3.2 Online Visualization: Module Selection and Online Button

Navigation tree for module selection:



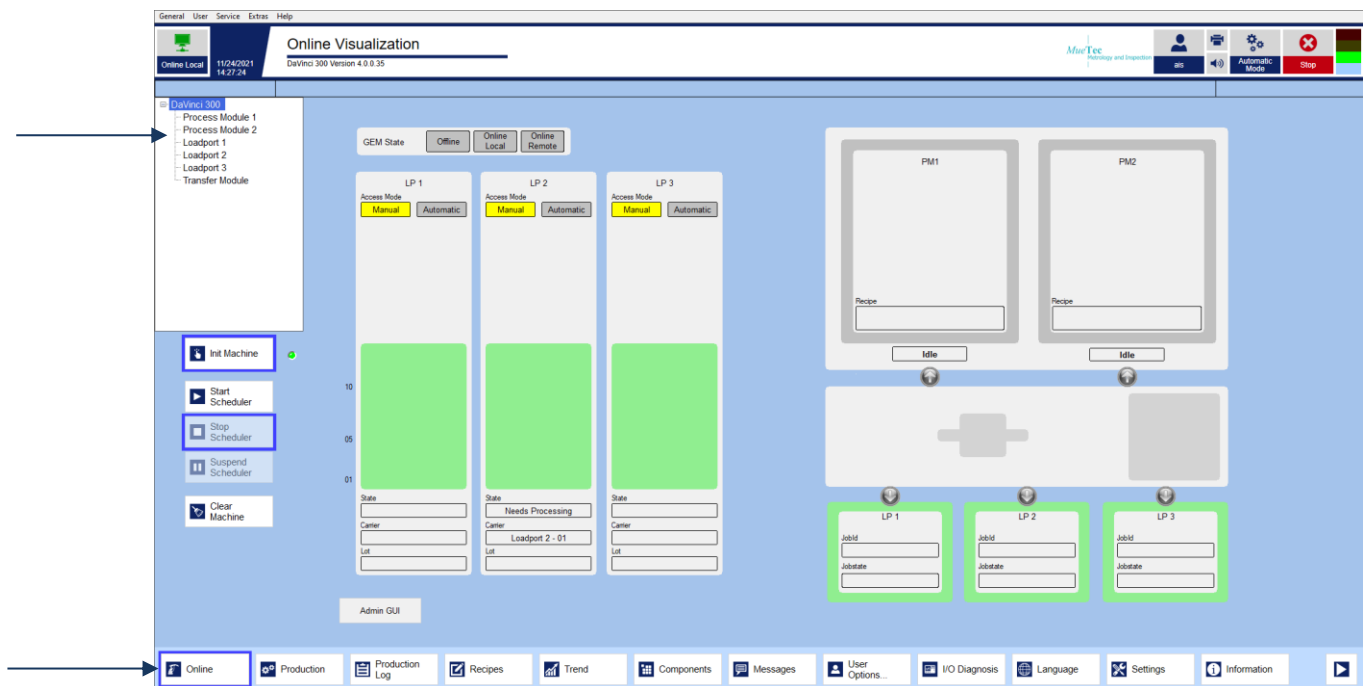
In the navigation tree the options DaVinci 200/300, Process Module 1 and 2, Loadport 1, 2 and 3 and the Transfer Module are available (depending on the customer's order).

You can only see the navigation tree if 'Online' is selected in the functions bar. You can check the selection of the button when 'Online' has a blue outlining like in the picture below.

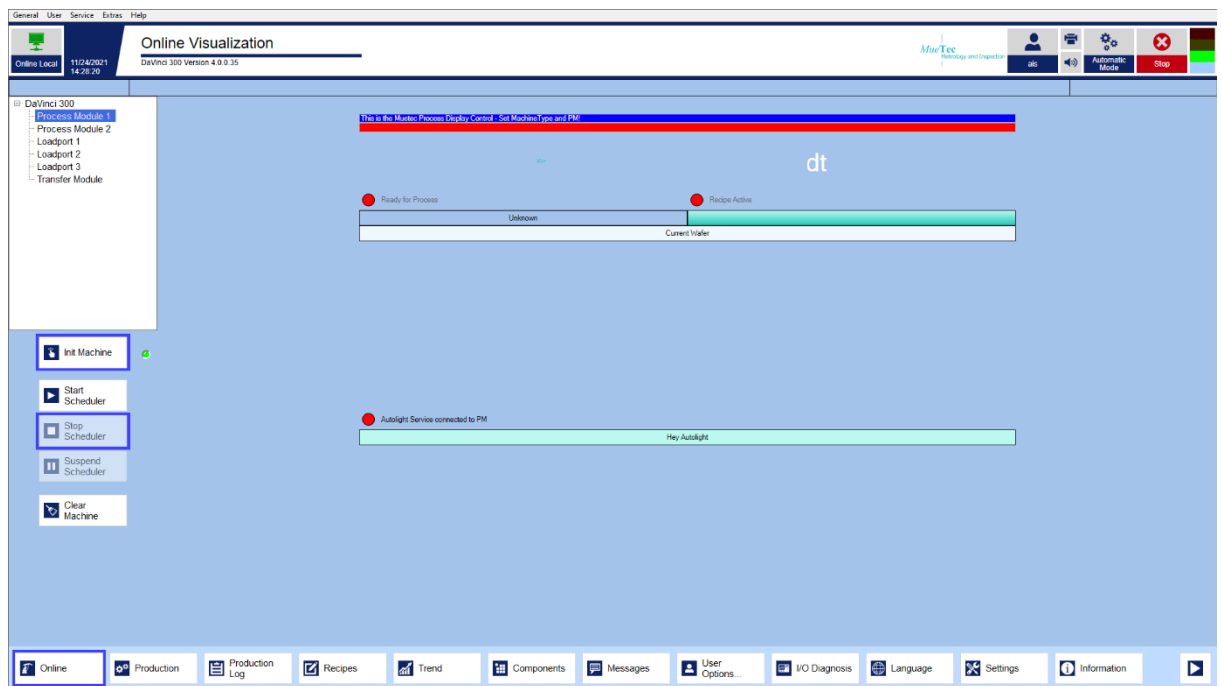


The following figure shows the overview screen with an explanation of the main components:

By selecting DaVinci 200/300 in the navigation tree and Online in the functions bar you will get the complete overview for your user level. Shown here is the operator's view:



If you choose Process Module1 or Process Module 2, you will get an overview of the activities, the process modules, and the current image acquisition as well as its status:



By clicking on load port 1 in the navigation tree, the status of the load port is displayed. When selecting load port 2 or load port 3 the same display appears for the respective load port.



By choosing Transfer Module in the navigation tree, the Transfer Module is shown. There you can see the robot as well as the prealigner and their status:



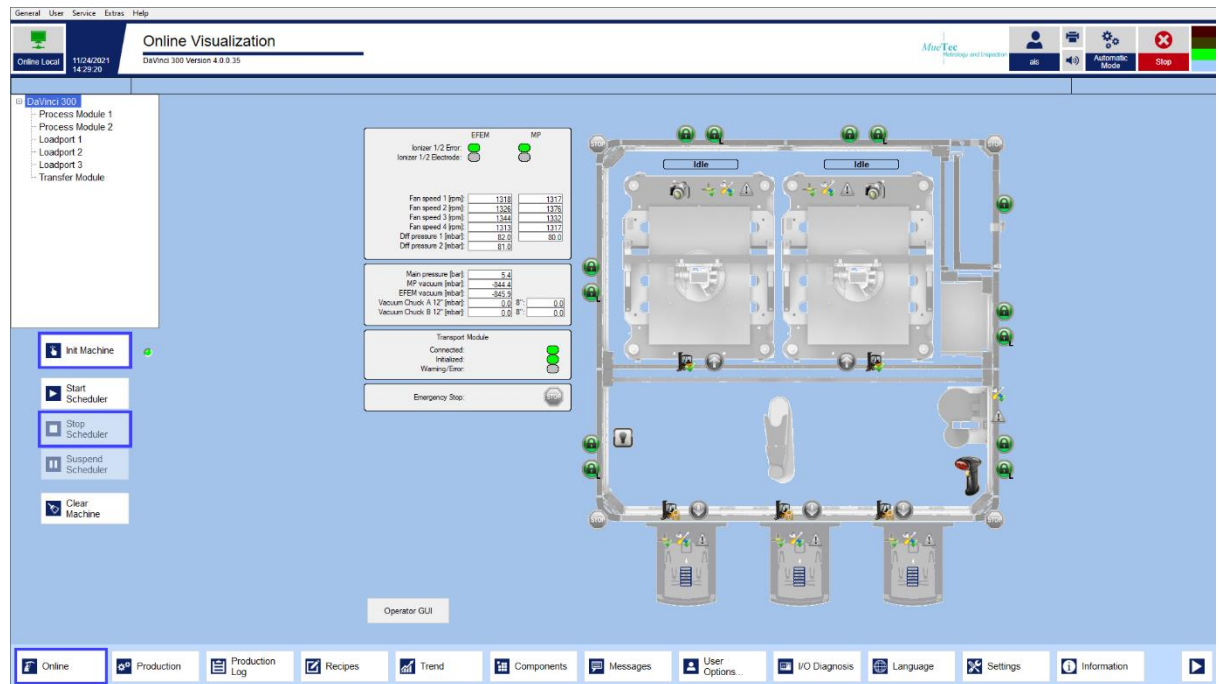
Please note:

Online must always be selected in the functions bar because the Visualization is only available in this view.

3.3.3 Admin GUI: Supervisor and/or Service Permissions

In the Supervisor / Service view a systematic overview of all modules is shown. Cassettes and substrates that are currently located at the load ports, the transfer module and/or in the process modules are represented.

The figure gives an example of a system overview for users, who are equipped with supervisor or service permissions:

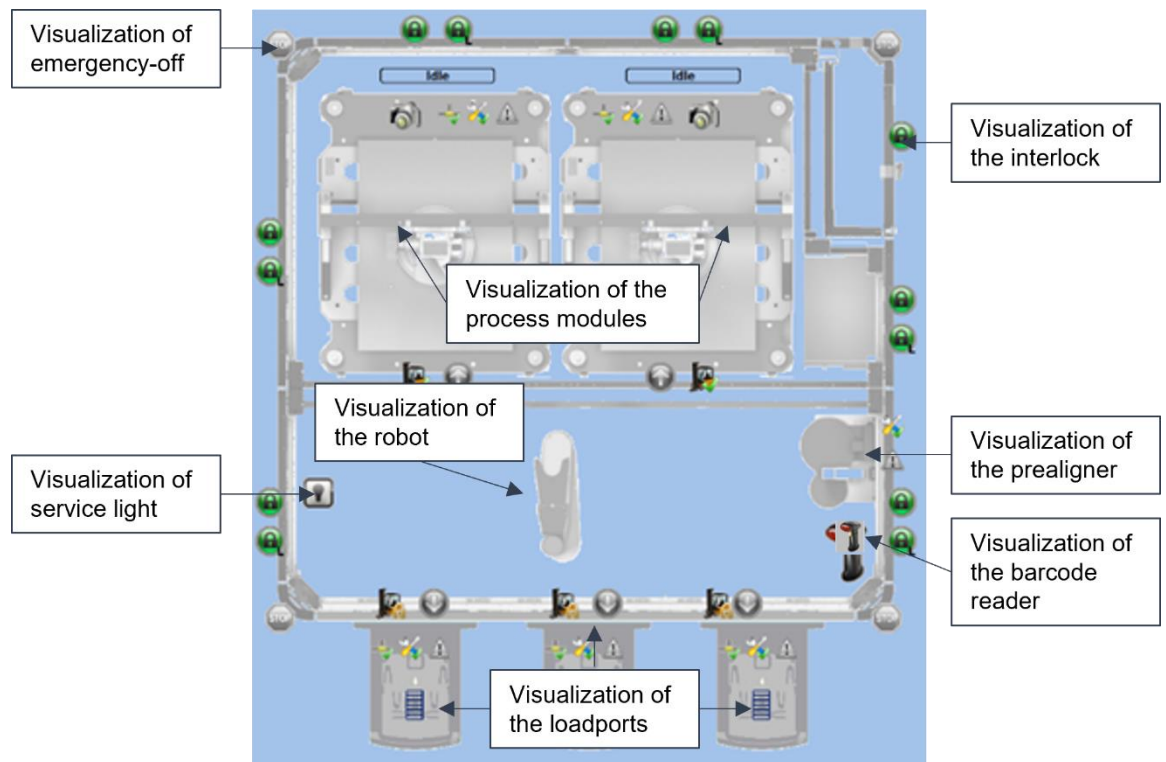


Details such as the fan speed and vacuum values are also given here. The symbols and coloration with respective meaning are explained in more detail later.

3.3.4 Visualization: System Overview

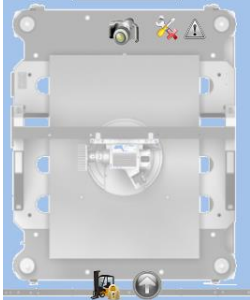
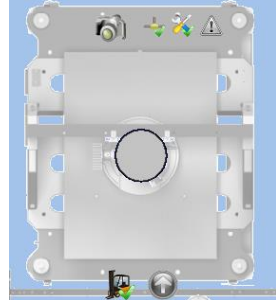
The system overview gives you a visualization of the system from the top. In this view you can monitor the status of the system.

For the supervisor and service, details of the system are shown, which are represented with special symbols. These symbols, the corresponding colors and their meanings are explained more detailed in the following.



Emergency stops and interlocks of the housing pannels and cassettes are marked in an easy way to simplify the display.

The main modules of the system overview include:

The process module: here the wafer is scanned	 Module: free	 Module: occupied by wafer
The robot: its movements and wafer allocations are displayed	 Robot in home position	 Robot transports a wafer
The prealigner: the wafer allocation and status are displayed		
The load port: the status is displayed		
The barcode reader: the status is displayed		
The cassette monitoring (Interlocks for housing panels): these are normally colored green - only colored red in case of an error (13 different instances)	 Interlock on cassette (housing panel) top	
	 Interlock on cassette (housing panel) bottom	
The emergency stop switch: these are normally colored grey - only red when triggered (4 instances, one at each corner)	 Emergency stop button	
Service light		

Other symbols are module-specific or only occur in the event of a failure.

Important symbols:

Emergency stop switch: there are four emergency stop switches on the system. If one is pressed, the system will stop. The status is displayed with two different symbols.		Normal operation, everything is fine, the system can be operated.		The emergency stop switch has been triggered at the indicated position. The problem must be solved before the system can be used again. Inform your supervisor.
Interlock switches on the cassettes: the housing has feedbacked in which state it is.		If all interlock signals are green, you can use the system.		One of the cassettes forming the housing has been removed, the position of the interlock is displayed in the system overview. The system cannot be operated until the problem is solved. Inform your supervisor.
Module / Component status: the symbol shows the current state of the module or component.		Normal operation, there are no errors and no warnings.		Warning, but no error. For example, if the module is not initialized, a warning will appear in the error display line. Acknowledge the warning. If necessary, initialize the module.
				Error! One of the modules indicates that there is a problem. An error message will appear in the error display line. Acknowledge the error and initialize the module if necessary. If necessary, inform your supervisor.
Initialization status: each main module, transport system (robot with prealigner), loadport and process module shows their initialization state with this symbol.		The main module is initialized, the component is ready.		The main module is not initialized. Initialize the corresponding main module. Use the Init button.
Inspection status: status of the process module		Idle The process module is ready and waiting for a wafer.		Active The process module is active and is processing a wafer.
				The process module has processed one wafer. Processing is complete.

Display of a loading/unloading request	 The module is ready for loading/unloading.	 The module is ready. A loading/unloading request has been sent and released.	 The module has received a loading/unloading request. However, this was refused.	
Wafer status in the system	 Wafer is in the module, but is unprocessed	 Wafer is currently being processed in the module.	 Wafer is processed.	 Wafer was processed, but an error occurred during processing.
Connection status of a module	 The module is connected.		 The module is not connected.	

Supervisor

If you have any questions or uncertainties, please contact your supervisor.

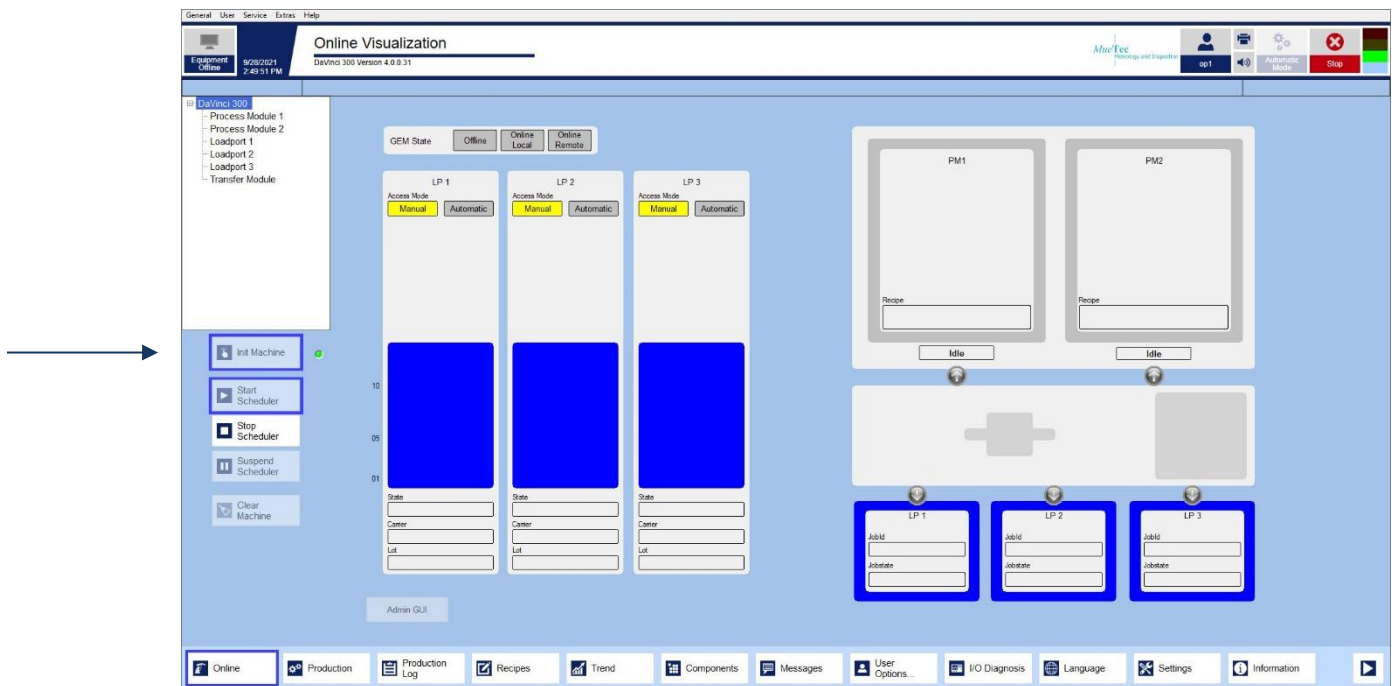
Service

If technical problems occur or if service tasks are pending, the supervisor must inform the service personnel.

3.3.5 Online Visualization: Operator GUI

The online visualization of the Operator GUI shows a systematic overview of all modules, selected components as well as cassettes and substrates that are currently located at the load ports, in the handling system and/or in the process modules. Only a short introduction is given here.

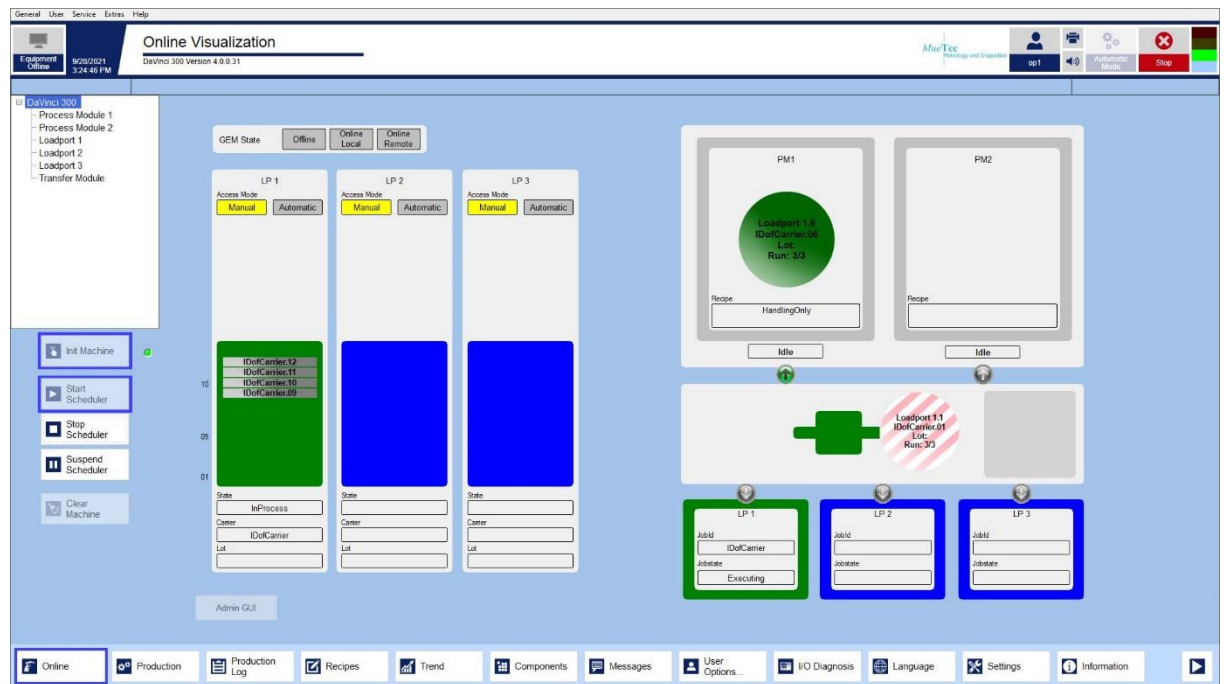
This figure shows the overview screen of the Online Visualization in the operator mode directly after the initialization of the system:



The system is initialized and ready for operation. There are no errors. The signal light is green, you can see this next to the Init Machine button. The modules' status is indicated by their coloring. This helps the operator to get a quick overview and diagnosis of the system and the module status. The modules can have the colors blue, green, yellow and red.

The exact reference of the colors per module is described in the following.

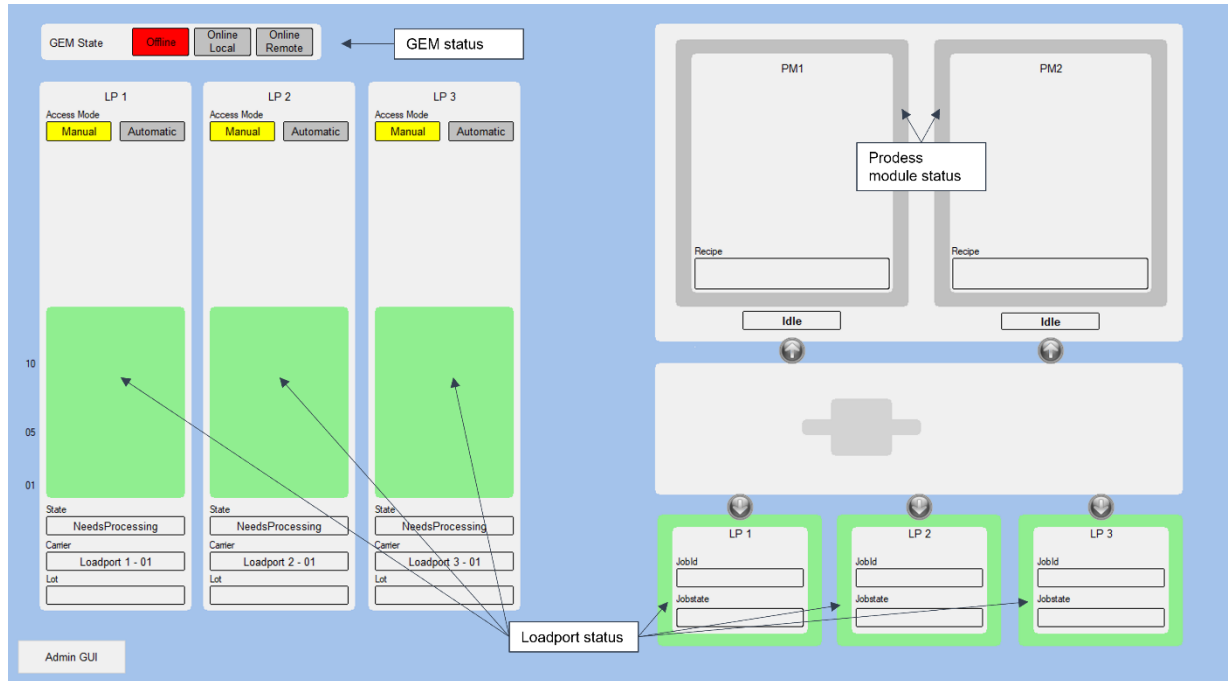
In this figure an example of the overview screen is shown during the operation for users with operator rights:



The main difference between the Operator GUI and Admin GUI is only the depiction in the screen area. While the screen area in the Admin GUI provides a detailed view of the system, the screen area in the Operator GUI allows you to quickly assess the overall situation.

3.3.6 Screen Area Operator GUI: GEM Status, System Overview, LP Overview

The system overview in the Operator mode serves as simplified view of the entire system in top view. Here you can monitor the current state of the system as well as the running jobs and the states of the wafers on the modules with coupled recipes.



GEM Status

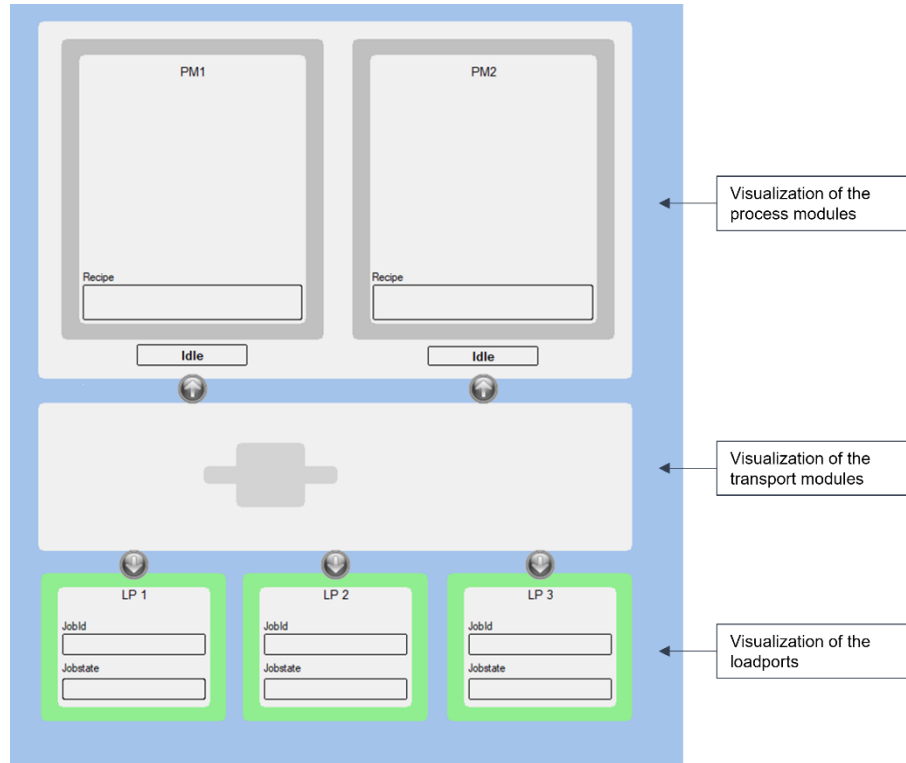


The status of the GEM connection can be **Offline**, **Online Local** and **Online Remote**. The respective active state is highlighted in color.

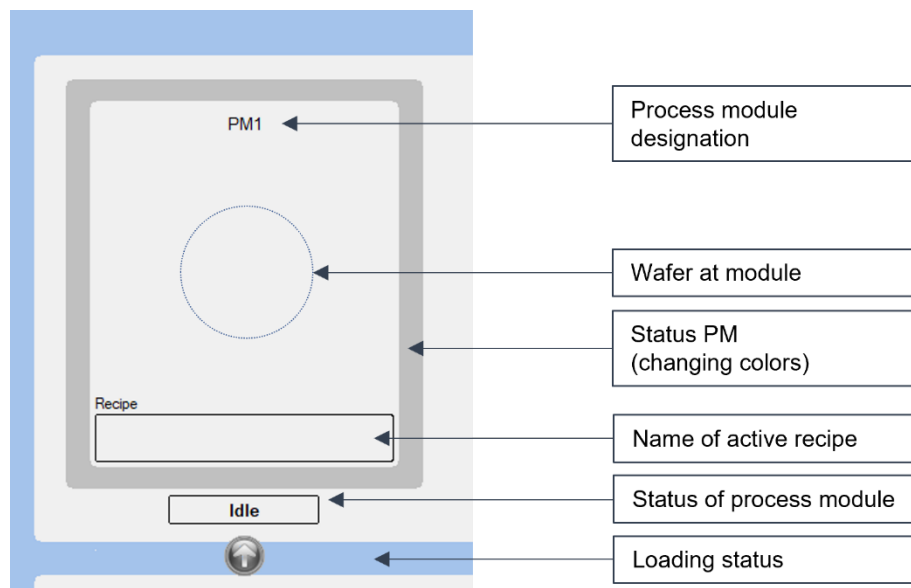
Status	Description
Offline	The host interface is disabled, or the host is offline. In this case, the background color of the Offline field is red. The background color of the Online Local and Online Remote fields is gray.
Online Local	The machine is in Online Local state if this is yellow. The host interface is enabled, and the host is listening and getting status and data of the tool. In this state, the background color of the Online Local field is yellow. The background color of the Offline and Online Remote fields is gray.
Online Remote	The machine is in Online Remote state if this is green. The host interface is enabled, and the tool is in automated mode. In this case, the background color of the Online Remote field is green. The background color of the Offline and Online Local fields is gray.

System Overview

The system overview in operator mode shows the three main components: Process modules (PM1 and PM2), transport module (robot and prealigner) and load ports (LP1, LP2 and LP3). They are presented in a simple manner to provide a quick overview.



Area for visualization of the process module:



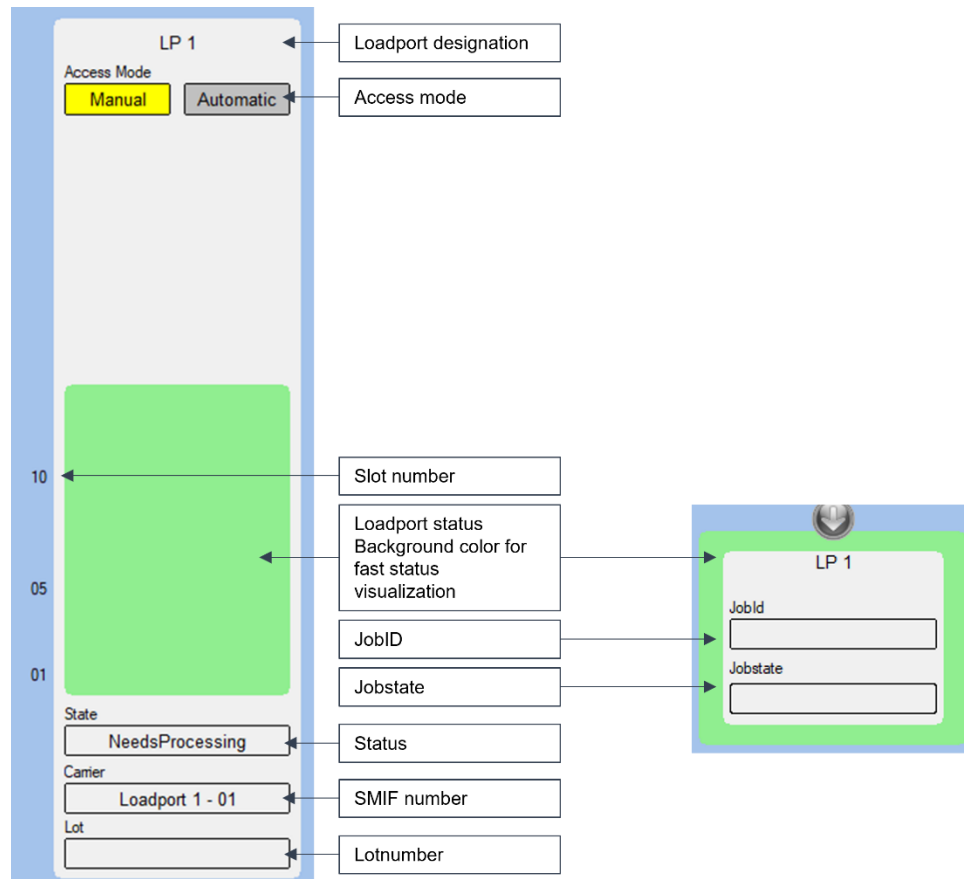
This display shows the process module and its status. The border shown here in grey is interactive and indicates the status of the PM via the color scheme. A wafer is placed on the PM if a circle pops up in the center of the PM visualization. The wafer status is indicated by

different coloring. In the window **Recipe** the respective recipe is displayed, which is the basis for the processing of the current wafer.

The status display field of the process module is mandatory and shows the states Idle (waiting, free), Active, Error or Init. These states can also be easily read by the color of the PM visualization window.

Load Port Overview

Load port visualization in operator view:



Information	Description	
Access mode	The background color of the fields Manual or Automatic shows the current active access mode of the load port.	
	The background color of Manual is yellow and of Automatic is grey.	The access mode is Manual.
	The background color of Manual is grey and of Automatic is yellow.	The access mode is Automatic.
Load port status	The current state of the load port module is indicated by the background color.	
	Grey	The load port module is in an undefined state, e.g., after a restart of the application without initialization.
	Red	The load port module is erroneous.
	Blue	The load port module is initialized, error-free and there is no carrier on the load port.
	Light green	The load port module is initialized and has no error. A carrier is on the load port, but the carrier is not docked.
	Green	The load port module is initialized and does not show errors. A carrier is on the load port. The carrier is docked.
Slot assignment	The current slot assignment is shown in the load port status information. After mapping a carrier, the current slot assignment is displayed in 25 fields. If a slot is occupied, the wafer number is displayed in the slot field. The background color of the slot field shows the material condition of the wafer in the slot.	
Status	The current processing status of the carrier is displayed in the status field load port information area. If no carrier is placed, this field is empty.	
Carrier	This field displays the ID read by the carrier ID reader. If no carrier is placed, this field is empty.	
Lot	This field shows the batch number assigned to the material of the carrier. If no batch is assigned to the material in the current carrier or no carrier is placed, the field is empty. In host "Online Remote" communication mode, batch information is sent from the host. If the host does not send the batch information, this field is blank. In other host communication modes, the batch ID can be entered in the Job Creator dialog.	

3.3.7 Substrate Information

An important source of information is the coloration of the wafer in the system. By different coloration of the wafers, you can quickly get an overview over the wafer status. The online visualization shows the colors of the wafers at the respective location/module.

State	Color, pattern	
UNDEFINED or without process/job		grey
NEEDS PROCESSING		magenta/white stripped
IN PROCESS		white
PROCESSED		dark green
REJECTED		orange
STOPPED		cyan
ABORTED		red
LOST		red/yellow stripped
SKIPPED		blue

4 Operation

This chapter describes how the operator works with the system. After a short overview, the important processes for the operator are described step by step. For details, see the *Software Operation Manual* and *TC User Documentation*.

4.1 Overview

The DaVinci 200 system is an automated metrology device for 200 mm (optional 150mm) wafers.

The inspection Cycle

Procedure of the system:

Setting Cassettes > (Optional Data Entry) > Inspection > (Optional Review of Inspection Results) > Removing Cassettes

We call each of these runs an *inspection cycle*.

The essential steps in a cycle are:

- The operator or the OHT places a FOUP/cassette/SMIF on the load port. The system recognizes the FOUP/cassette/SMIF and waits for your input (in fully Automatic Mode this step is omitted): **Load**. The system reads the RFID code and identifies the number of contained wafers (wafer mapping).
- Only in Online Local mode and Offline Mode (in fully automatic mode this step is omitted):
 - The operator selects a recipe in the Job Creator and determines which wafers should be inspected.
 - The operator presses **Start Job** on the touchscreen to start the inspection. The system now carries out the inspection.
- The operator or OHT removes the FOUP/cassette/SMIF.

The system must be switched on and operational when the operator starts a cycle.

Turning the system on and off, supplying power and vacuum and servicing the system are not the responsibility of the operator and are not described in this document.

4.2 Modes of Operation

The system offers three working modes:

- Offline Mode – Wafer data and inspection recipes are entered by the operator via the touchscreen. The inspection results are only available in the system.
- Online Local Mode – Wafer data and inspection recipes are entered by the operator via the touchscreen. The inspection results are automatically sent to a host computer.
- Online Remote Mode – Wafer data and inspection recipes are automatically read and send from a host computer.

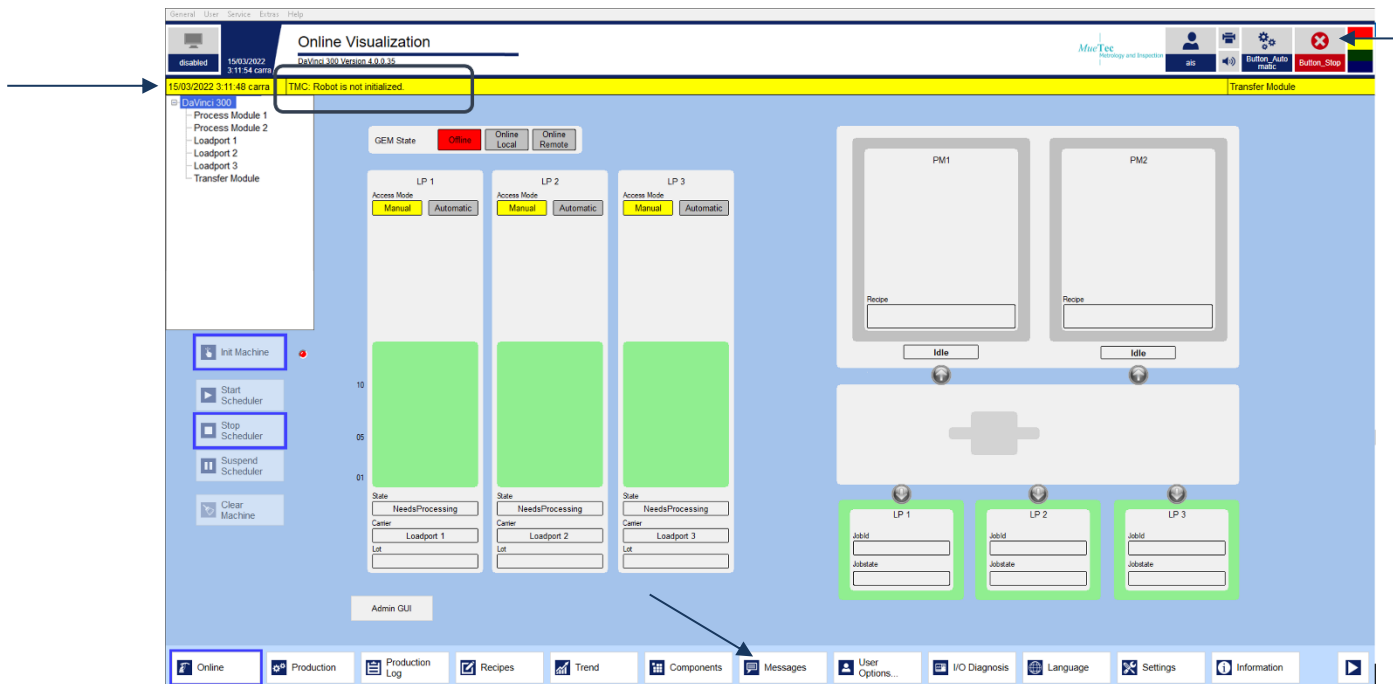
Apart from entering the wafer data, the inspection cycle is identical in the three modes.

4.3 System Messages

System messages that require the operator's attention are displayed on the touchscreen. In most cases, the status of the system is displayed, and an instruction or help may be given. If error messages or warnings appear that require the intervention of an expert, please inform your supervisor (e.g. if the vacuum fails).

Messages related to system's status may also appear in the status line at the top of the touch panel.

Possible areas in which messages can appear:



The error message can be found in the messages area. In case of an error, please contact your supervisor.

4.4 Starting the Software and Initializing the System

If the software is not yet initialized or the user interface is not ready when you step to the device, please proceed as follows.

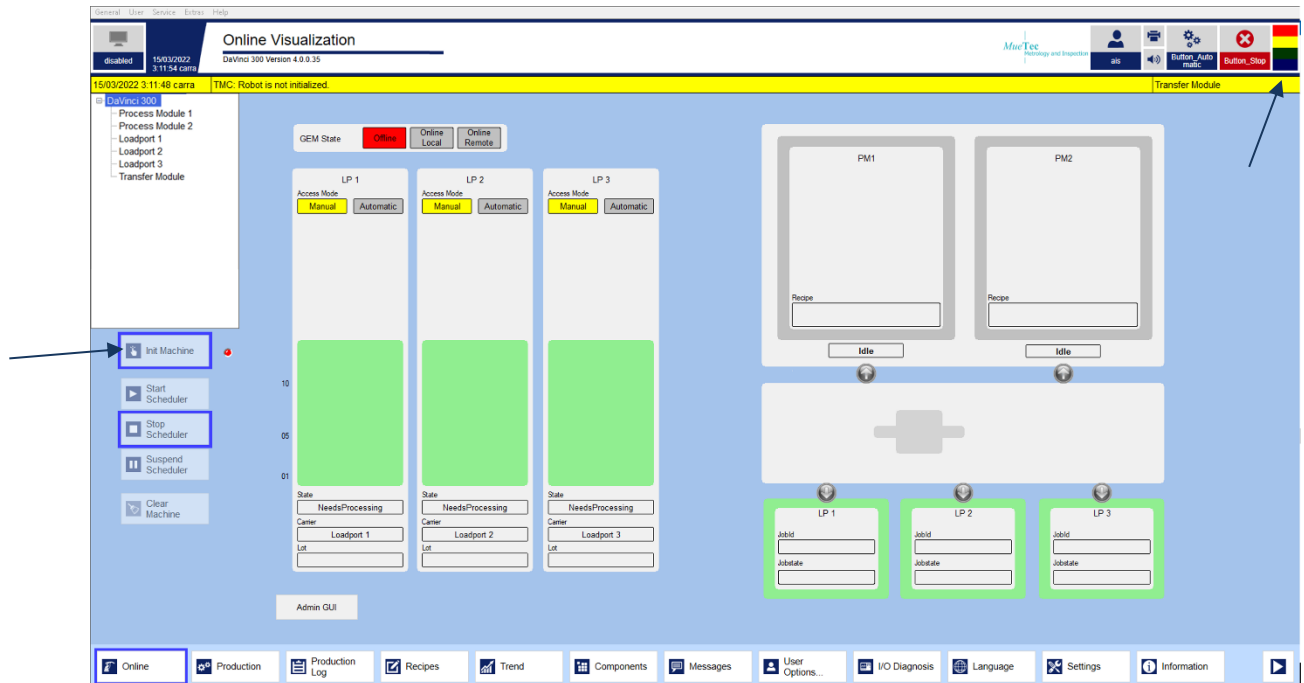
Start the software by clicking on the AIS ToolCommander symbol:



When the application is started for the first time, the system is automatically initialized after logging in with a username and password. Customer-dependent usernames can have the right for "automatic initialization".

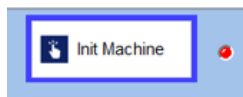
If you log in with a different username that is not assigned an automatic initialization, proceed as follows:

The visualization of the components is loaded. This can take a short moment. The visualization of the DaVinci 200 is started, the graphical user interface looks as follows:



In the figure above you can see the status display in the upper right corner. The system is not initialized. In the middle-left corner, you can see the button **Init Machine**. The red light indicates that the machine requires initialization. The signal light is yellow.

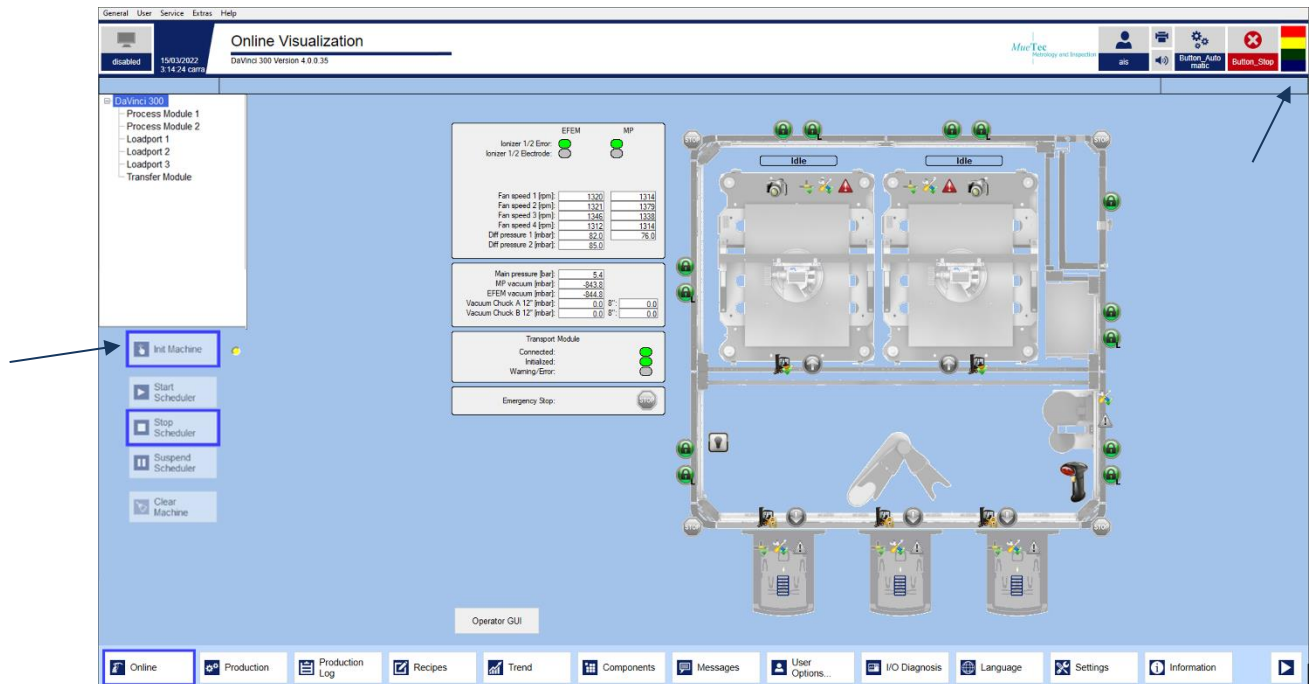
The **Init Machine** button indicates that the initialization has not been completed:



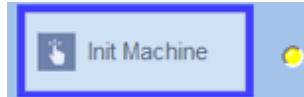
If the signal light is yellow, the upper status line shows: "Robot is not initialized" in yellow.

The robot or transport module (TM) is the first component of the system to be initialized. This indicates that the entire system requires initialization.

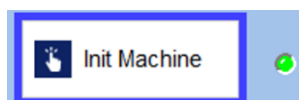
After pressing **Init Machine**, the initialization of the individual components begins:



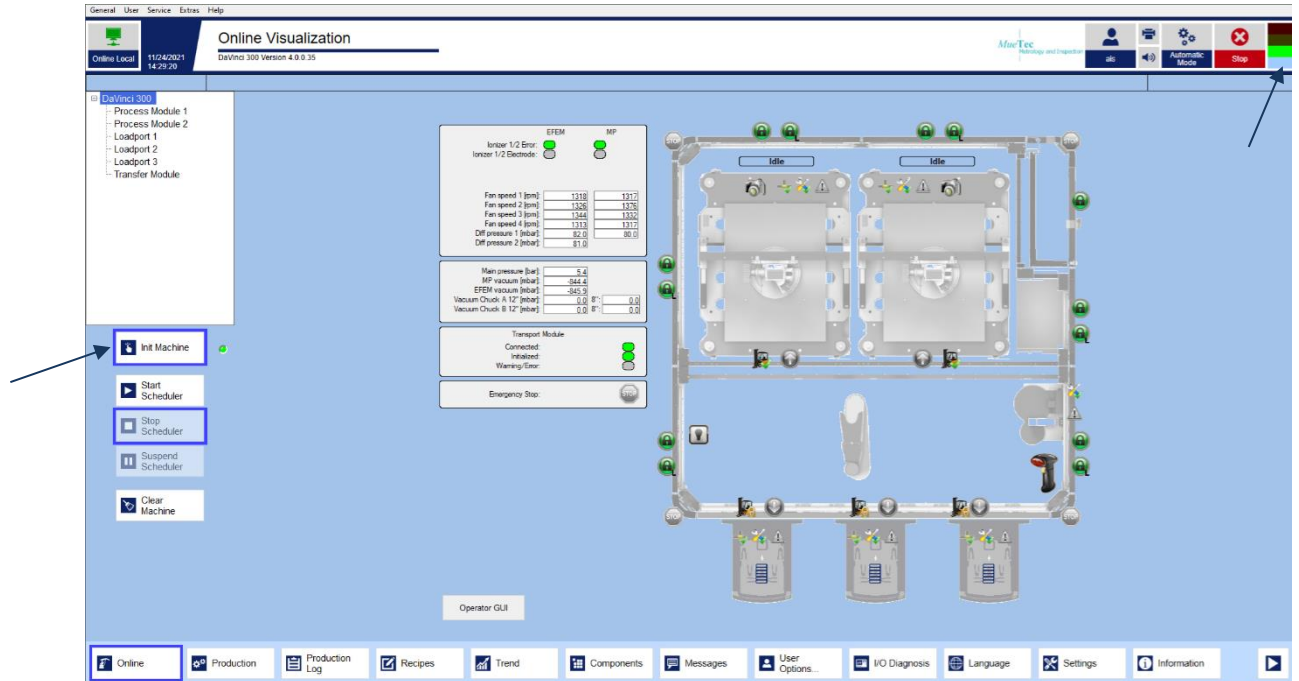
The yellow signal next to the **Init** Button and the grey **Init Machine** Button indicate that the initialization is in progress. The blue box flashes:



The green signal next to the Init button indicates that the initialization has been completed and the tool is now waiting for your input (the status line is also empty):



If the machine runs without errors, the surface looks like this:



The system is fully initialized and ready for use:

- The signal next to the **Init** button is green.
- The status bar shows no errors.
- The signal light is green.
- The system is now initialized and waits for further operating steps.

If the system has problems during initialization, you can recognize this by an error message in the status bar and by the red signal at the signal light. In this case, please contact a supervisor and do not proceed.

4.5 Setting Up Cassettes and Starting a Job

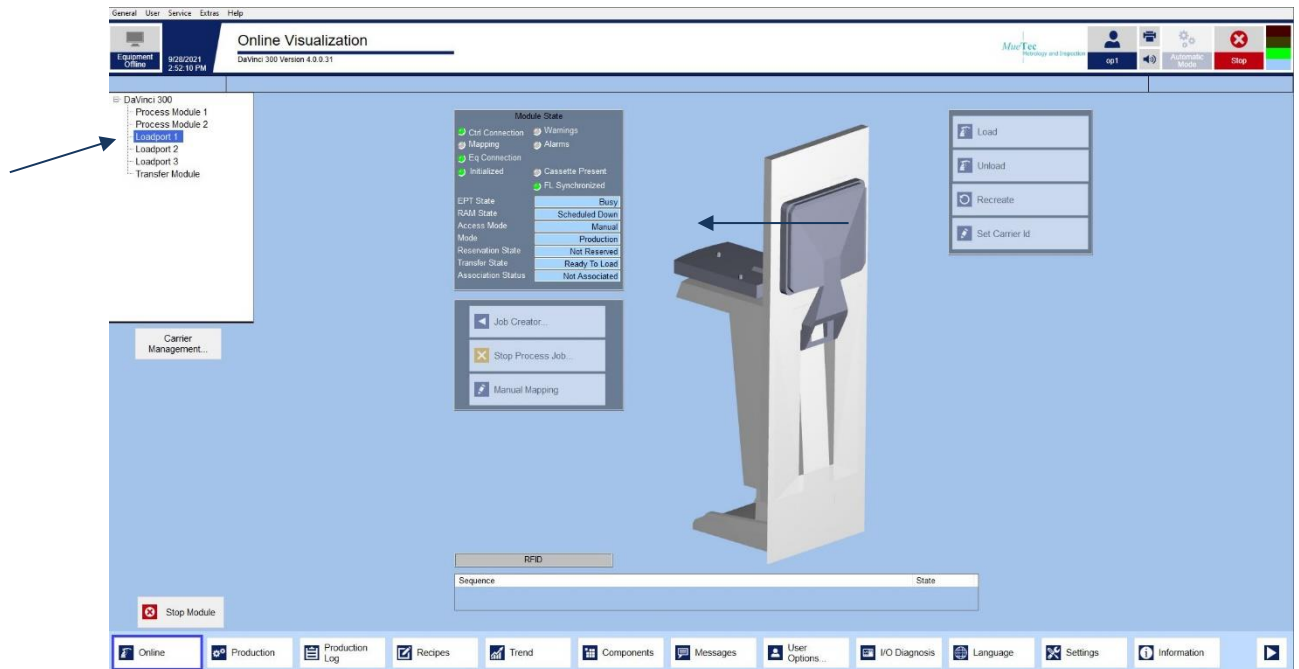
You can start an inspection cycle if the following prerequisites are met.

- The system is switched on and ready for operation. If this is not the case, contact your supervisor.
- No error messages are displayed.
- There are no wafers in the system.

If there are still wafers on the stations, determine the owner and their purpose and remove them. If in doubt, call your supervisor.

You can start a job if there are no error messages.

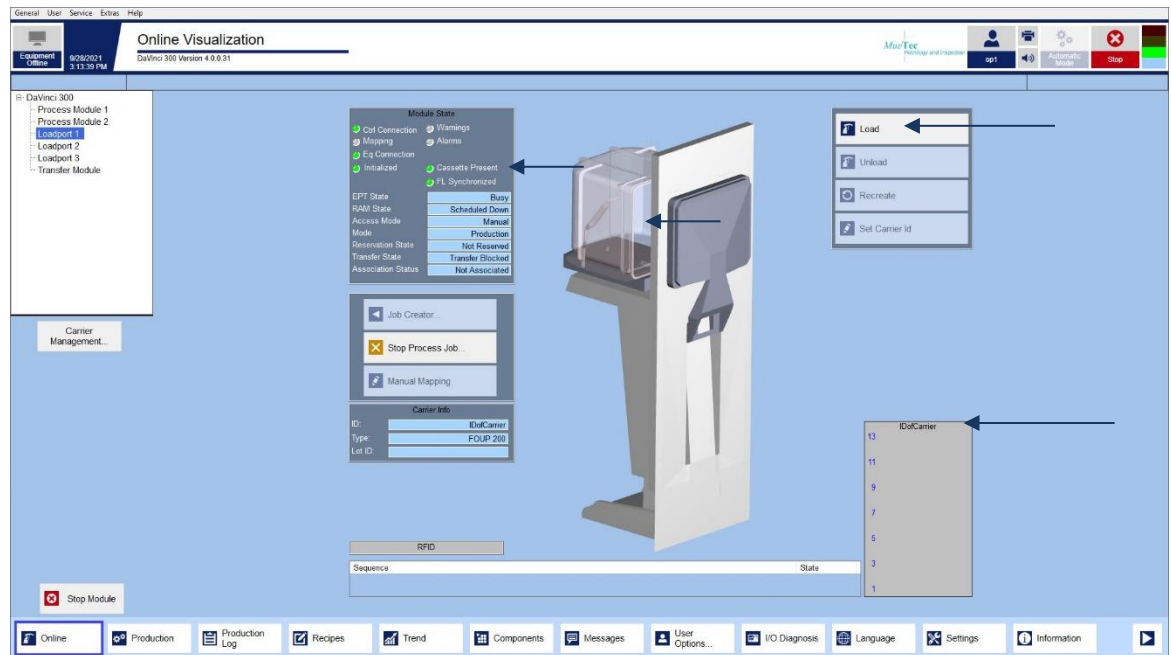
- Go to load port 1 in the component overview.
- Load port 1 signals its operational readiness (unloaded) and shows no errors (these two steps apply equally to load port 2 and 3).



- Place a FOUP/cassette/SMIF equipped with 200 mm (150mm optional) wafers on the free load port and wait a short moment. The load port detects the now loaded FOUP/cassette/SMIF.
- The RFID code is read (optional).
- You will see in the visualization that the FOUP/cassette/SMIF you have set up has been recognized and is ready for loading.

The **Load** button is not released until a FOUP/cassette/SMIF has been detected on the load port.

Representation of the load port in loaded state:



When the load port is loaded:

- Press the **Load** button.
- The FOUP/cassette/SMIF is now loaded automatically:
 - The FOUP/cassette/SMIF is docked.
 - The door of the FOUP/cassette/SMIF is then opened.
 - The next process step is the mapping of the FOUP/cassette/SMIF or the wafers.
- If no error messages are displayed, the Job Creator menu opens (optionally you must enter a CarrierID before the Job Creator opens).

If you receive an error message, there is a high probability that the mapping process was not ok. This may be due to the following reasons:

- Tilted FOUP/cassette/SMIF (if this is the case, call your supervisor or service personal)
- Tilted allocation of wafers in the FOUP/cassette/SMIF (cross slot)
- Two wafers in one slot (double slot)

In this case, proceed as follows:

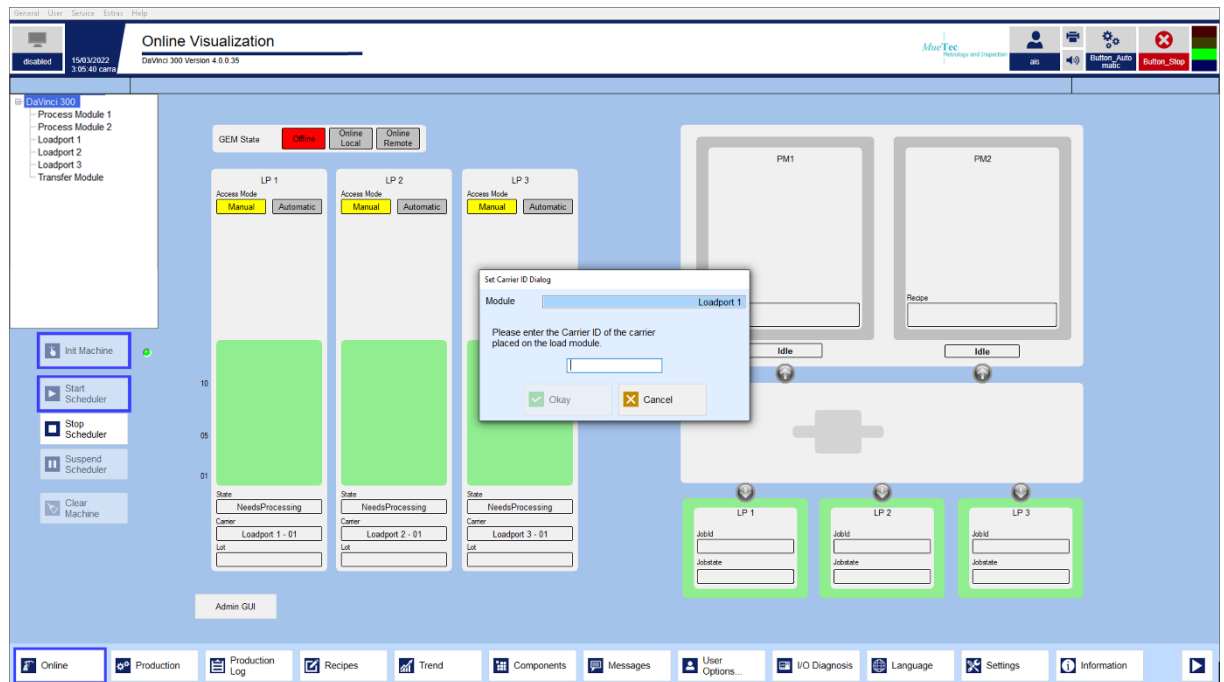
- In this case, unload the FOUP/cassette/SMIF with the **Unload** button or go to the Message menu item to acknowledge the error.
- Then correct the cause of the error in the FOUP/cassette/SMIF. If you are not authorized to do so, contact your supervisor.

After correcting the error, set up the FOUP/cassette/SMIF again and repeat the loading procedure.

A correctly loaded FOUP/cassette/SMIF is the basis for the correct process. Failure to observe the error messages or improper assembly can lead to destruction of the system components and the wafers in the FOUP/cassette/SMIF.

4.6 Loading FOUP/Cassette/SMIF and Setting CarrierID in GEM Mode Offline and Local Online

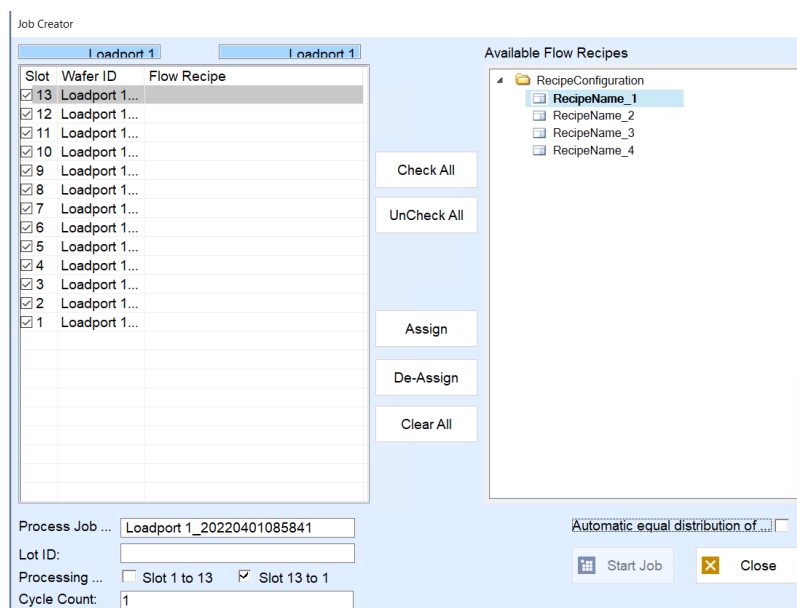
After loading a FOUP/cassette/SMIF in automatic or in manual mode, you must set the CarrierID (in GEM Mode Offline and Local Online):



After adding the CarrierID, click the OK button.

Depending on your host system there are different options for the load port settings.

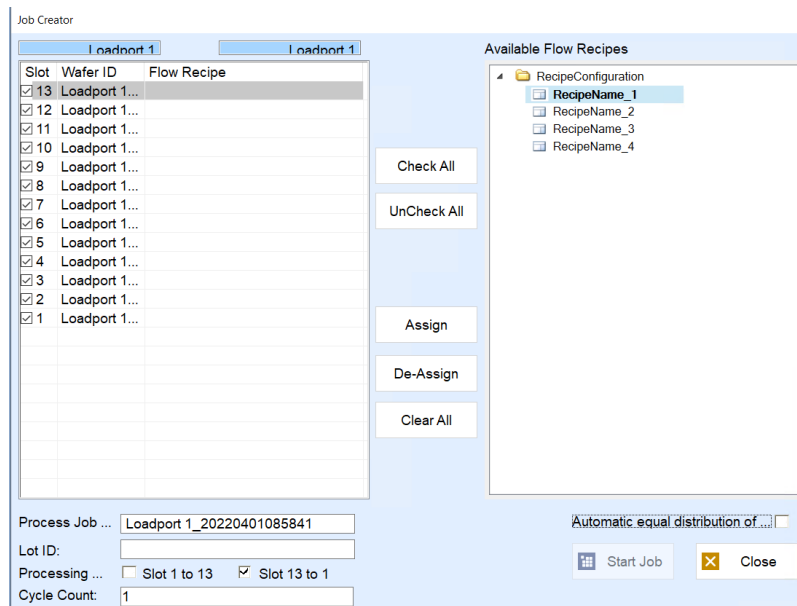
If the option **autoload after place** is activated in the base config of the load port and after setting the CarrierID, the Job Creator window appears automatically.



If the option **autoload after place** is **not** activated in the base config of the Loadport and after setting the CarrierID, you must load the FOUP/cassette/SMIF by clicking the load button in this view.



After loading the FOUP/cassette/SMIF by clicking the load button the Job Creator will open automatically:



If you closed the Job Creator, click the button Job Creator and the Job Creator window appears.

4.7 Job Creator and Start of a Measurement Cycle

Job Creator

Slot	Wafer ID	Flow Recipe
☒ 13	Loadport 1...	
☒ 12	Loadport 1...	
☒ 11	Loadport 1...	
☒ 10	Loadport 1...	
☒ 9	Loadport 1...	
☒ 8	Loadport 1...	
☒ 7	Loadport 1...	
☒ 6	Loadport 1...	
☒ 5	Loadport 1...	
☒ 4	Loadport 1...	
☒ 3	Loadport 1...	
☒ 2	Loadport 1...	
☒ 1	Loadport 1...	

Check All
UnCheck All
Assign
De-Assign
Clear All

Available Flow Recipes

- RecipeConfiguration
 - RecipeName_1
 - RecipeName_2
 - RecipeName_3
 - RecipeName_4

Process Job ... Loadport 1_20220401085841

Lot ID:

Processing ... ☐ Slot 1 to 13 ☒ Slot 13 to 1

Cycle Count: 1

Automatic equal distribution of ... ☐

Start Job Close

On the left you can see the wafers found during mapping, on the right the available recipes. All mapped wafers are selected by default. The task of the Job Editor is to link material and recipe. The Job Creator enables you to link all wafers with recipes from a correctly assigned FOUP/cassette/SMIF by default setting.

The meaning and usage of the different buttons will be explained in the following.

Check All

With **Check All**, all mapped wafers are selected:

Job Creator

Slot	Wafer ID	Flow Recipe
☒	13 Loadport 1...	
☒	12 Loadport 1...	
☒	11 Loadport 1...	
☒	10 Loadport 1...	
☒	9 Loadport 1...	
☒	8 Loadport 1...	
☒	7 Loadport 1...	
☒	6 Loadport 1...	
☒	5 Loadport 1...	
☒	4 Loadport 1...	
☒	3 Loadport 1...	
☒	2 Loadport 1...	
☒	1 Loadport 1...	

Check All
UnCheck All
Assign
De-Assign
Clear All

Available Flow Recipes

- RecipeConfiguration
 - RecipeName_1**
 - RecipeName_2
 - RecipeName_3
 - RecipeName_4

Process Job ... Loadport 1_20220401085841
Lot ID:
Processing ... ☐ Slot 1 to 13 ☒ Slot 13 to 1
Cycle Count: 1

Automatic equal distribution of ... ☐

Start Job Close

UnCheck All

By using **UnCheck All** no wafer is selected anymore:

Job Creator

Slot	Wafer ID	Flow Recipe
☐	13 Loadport 1...	
☐	12 Loadport 1...	
☐	11 Loadport 1...	
☐	10 Loadport 1...	
☐	9 Loadport 1...	
☐	8 Loadport 1...	
☐	7 Loadport 1...	
☐	6 Loadport 1...	
☐	5 Loadport 1...	
☐	4 Loadport 1...	
☐	3 Loadport 1...	
☐	2 Loadport 1...	
☐	1 Loadport 1...	

Check All
UnCheck All
Assign
De-Assign
Clear All

Available Flow Recipes

- RecipeConfiguration
 - RecipeName_1**
 - RecipeName_2
 - RecipeName_3
 - RecipeName_4

Process Job ... Loadport 1_20220401085841
Lot ID:
Processing ... ☐ Slot 1 to 13 ☒ Slot 13 to 1
Cycle Count: 1

Automatic equal distribution of ... ☐

Start Job Close

Assign

The **Assign** button allocates the chosen recipe to the checked wafers:

The screenshot shows the 'Job Creator' window. On the left, a table lists slots and wafers. Slot 13 is checked, and its wafer ID is 'Loadport 1...'. The flow recipe assigned to it is 'RecipeName_1'. The 'Assign' button is highlighted. On the right, the 'Available Flow Recipes' list shows 'RecipeName_1' selected. Below the table, there are buttons for 'Check All', 'UnCheck All', 'Assign', 'De-Assign', and 'Clear All'. At the bottom, there are input fields for 'Process Job ...' (Loadport 1_20220401085841), 'Lot ID:', 'Processing ...' (Slot 1 to 13, Slot 13 to 1), and 'Cycle Count: 1'. There are also 'Start Job' and 'Close' buttons.

Slot	Wafer ID	Flow Recipe
☐ 13	Loadport 1...	RecipeName_1
☐ 12	Loadport 1...	RecipeName_1
☐ 11	Loadport 1...	RecipeName_1
☐ 10	Loadport 1...	RecipeName_1
☐ 9	Loadport 1...	RecipeName_1
☐ 8	Loadport 1...	RecipeName_1
☐ 7	Loadport 1...	RecipeName_1
☐ 6	Loadport 1...	RecipeName_1
☐ 5	Loadport 1...	RecipeName_1
☐ 4	Loadport 1...	RecipeName_1
☐ 3	Loadport 1...	RecipeName_1
☐ 2	Loadport 1...	RecipeName_1
☐ 1	Loadport 1...	RecipeName_1

Available Flow Recipes

- RecipeConfiguration
 - ☒ RecipeName_1
 - ☐ RecipeName_2
 - ☐ RecipeName_3
 - ☐ RecipeName_4

Buttons: Check All, UnCheck All, Assign, De-Assign, Clear All

Process Job ...: Loadport 1_20220401085841

Lot ID:

Processing ...: ☐ Slot 1 to 13 ☒ Slot 13 to 1

Cycle Count: 1

Automatic equal distribution of ... ☐

Buttons: Start Job, Close

An error appears if there are no wafers checked to which a recipe can be assigned:

The screenshot shows the 'Job Creator' window with an error message. In the table, Slot 13 is checked, but its wafer ID is 'Loadport 1...' and the flow recipe is 'RecipeName_2'. The 'Assign' button is highlighted. The error message 'Invalid Selection, No Wafer Exists' is displayed in red. The 'Available Flow Recipes' list shows 'RecipeName_2' selected. Below the table, there are buttons for 'Check All', 'UnCheck All', 'Assign', 'De-Assign', and 'Clear All'. At the bottom, there are input fields for 'Process Job ...' (Loadport 1_20220401085841), 'Lot ID:', 'Processing ...' (Slot 1 to 13, Slot 13 to 1), and 'Cycle Count: 1'. There are also 'Start Job' and 'Close' buttons.

Slot	Wafer ID	Flow Recipe
☒ 13	Loadport 1...	RecipeName_2
☐ 12	Loadport 1...	RecipeName_2
☐ 11	Loadport 1...	RecipeName_2
☐ 10	Loadport 1...	RecipeName_2
☐ 9	Loadport 1...	RecipeName_2
☐ 8	Loadport 1...	RecipeName_2
☐ 7	Loadport 1...	RecipeName_2
☐ 6	Loadport 1...	RecipeName_2
☐ 5	Loadport 1...	RecipeName_2
☐ 4	Loadport 1...	RecipeName_2
☐ 3	Loadport 1...	RecipeName_2
☐ 2	Loadport 1...	RecipeName_2
☐ 1	Loadport 1...	RecipeName_2

Available Flow Recipes

- RecipeConfiguration
 - ☐ RecipeName_1
 - ☒ RecipeName_2
 - ☐ RecipeName_3
 - ☐ RecipeName_4

Buttons: Check All, UnCheck All, Assign, De-Assign, Clear All

Invalid Selection, No Wafer Exists

Process Job ...: Loadport 1_20220401085841

Lot ID:

Processing ...: ☐ Slot 1 to 13 ☒ Slot 13 to 1

Cycle Count: 1

Automatic equal distribution of ... ☐

Buttons: Start Job, Close

De-Assign

With the **De-Assign** button, no recipe is assigned to the selected wafers anymore:

Job Creator

Slot	Wafer ID	Flow Recipe
☐ 13	Loadport 1...	RecipeName_1
☐ 12	Loadport 1...	RecipeName_1
☐ 11	Loadport 1...	RecipeName_1
☐ 10	Loadport 1...	RecipeName_1
☐ 9	Loadport 1...	
☐ 8	Loadport 1...	RecipeName_1
☐ 7	Loadport 1...	
☐ 6	Loadport 1...	RecipeName_1
☐ 5	Loadport 1...	
☐ 4	Loadport 1...	RecipeName_1
☐ 3	Loadport 1...	RecipeName_1
☐ 2	Loadport 1...	RecipeName_1
☐ 1	Loadport 1...	RecipeName_1

Check All
UnCheck All
Assign
De-Assign
Clear All

Available Flow Recipes

- RecipeConfiguration
 - RecipeName_1
 - RecipeName_2
 - RecipeName_3
 - RecipeName_4

Process Job ... Loadport 1_20220401085841
Lot ID:
Processing ... ☐ Slot 1 to 13 ☒ Slot 13 to 1
Cycle Count: 1

Automatic equal distribution of ... ☐

Start Job Close

Clear All

With **Clear All**, all previously made settings are reset.

Job Creator

Slot	Wafer ID	Flow Recipe
☐ 13	Loadport 1...	
☐ 12	Loadport 1...	
☐ 11	Loadport 1...	
☐ 10	Loadport 1...	
☐ 9	Loadport 1...	
☐ 8	Loadport 1...	
☐ 7	Loadport 1...	
☐ 6	Loadport 1...	
☐ 5	Loadport 1...	
☐ 4	Loadport 1...	
☐ 3	Loadport 1...	
☐ 2	Loadport 1...	
☐ 1	Loadport 1...	

Check All
UnCheck All
Assign
De-Assign
Clear All

Available Flow Recipes

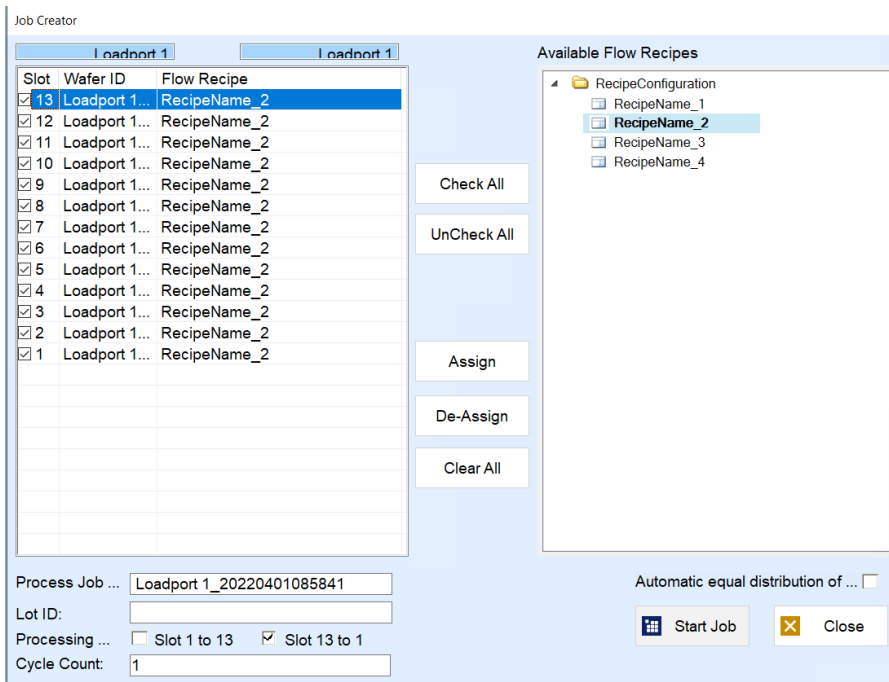
- RecipeConfiguration
 - RecipeName_1
 - RecipeName_2
 - RecipeName_3
 - RecipeName_4

Process Job ... Loadport 1_20220401085841
Lot ID:
Processing ... ☐ Slot 1 to 13 ☒ Slot 13 to 1
Cycle Count: 1

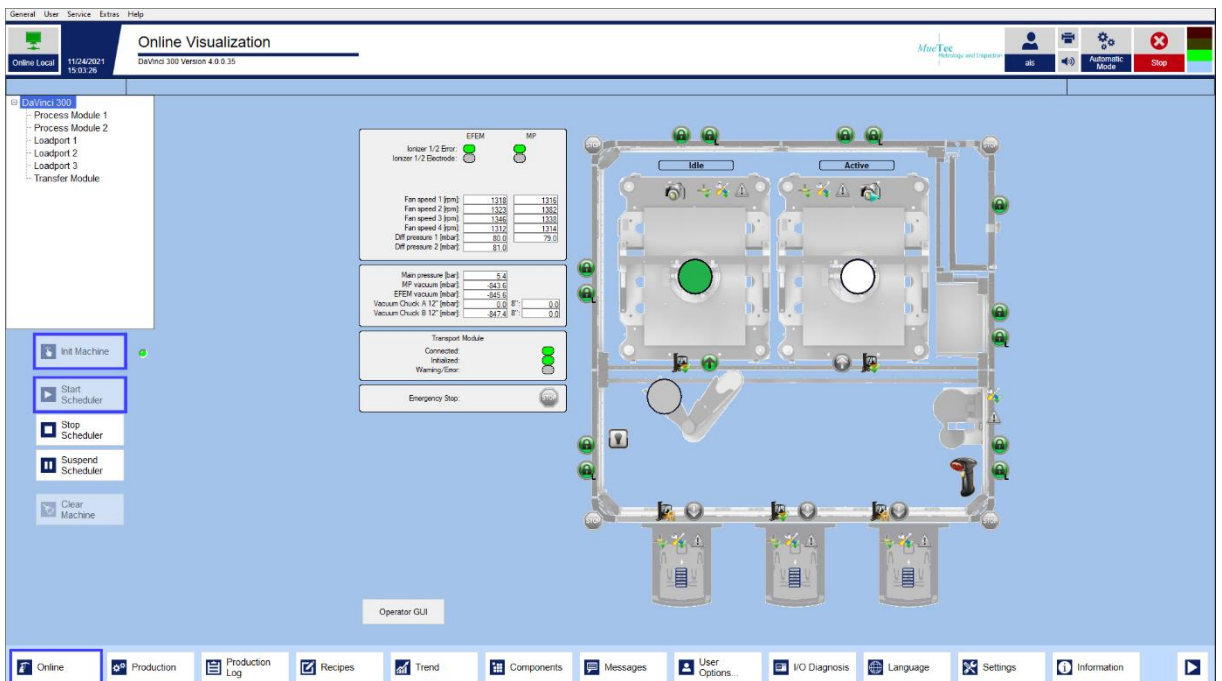
Automatic equal distribution of ... ☐

Start Job Close

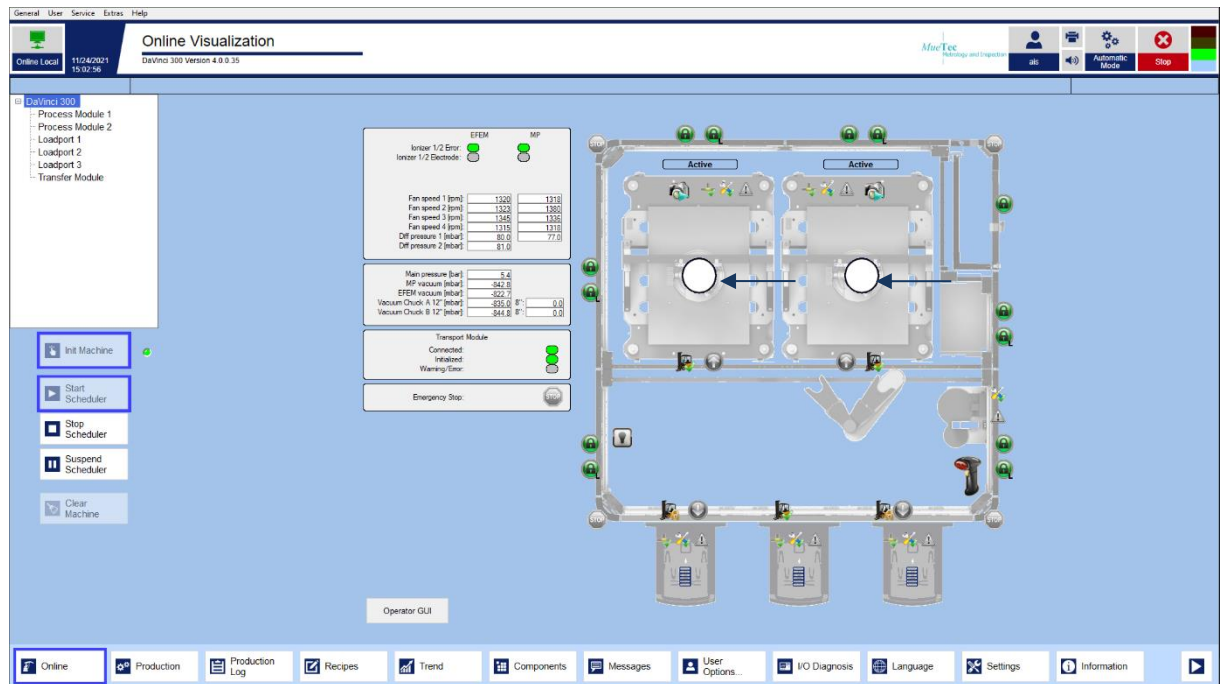
After selecting the desired wafers and recipes, the measurement process can be started with clicking the Start Job button.



The system will process the selected wafers with the given recipe. The status of the wafer in the system is displayed in the online visualization:



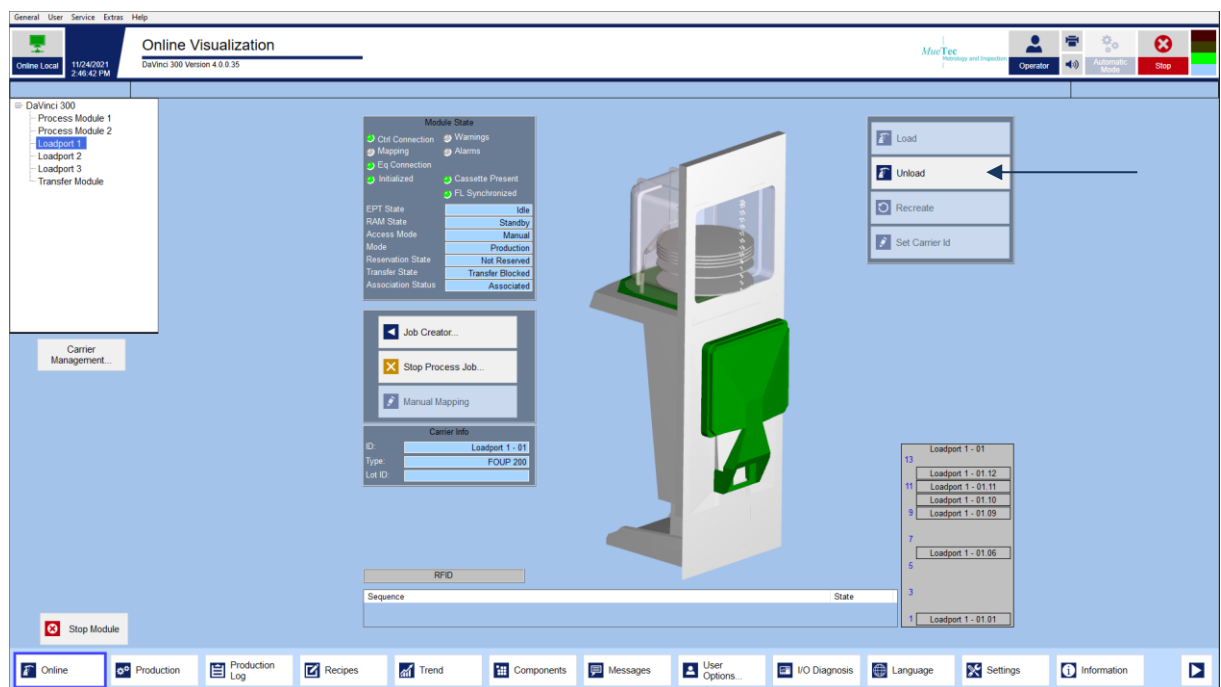
This picture shows the process "wafer on measuring platform":



The system always informs you of the process flow. A green wafer depiction means that the process is complete.

When processing is complete, you can unload the FOUP/cassette/SMIF. If the system is in the mode Online Local or Online Remote the FOUP/cassette/SMIF will be unloaded automatically if the option is active. The status "Done" is displayed in which all wafers in the load port are highlighted in green.

With the command **Unload** the FOUP/cassette/SMIF can be unloaded from the load port:



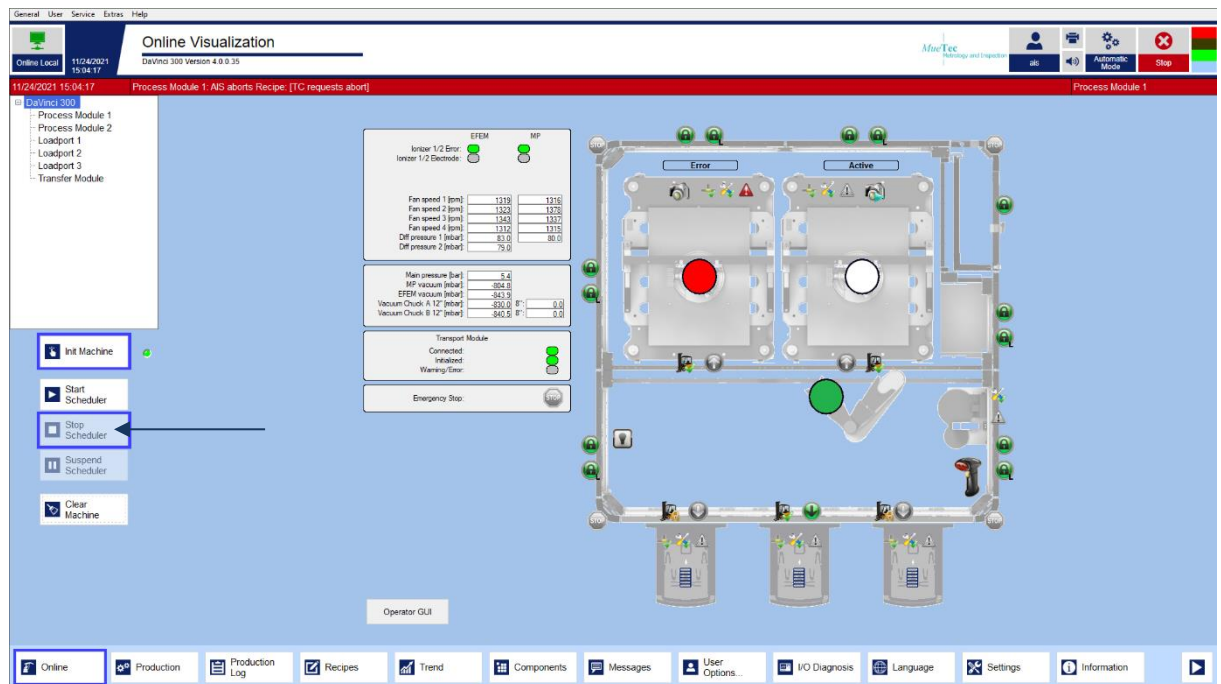
Unload of the FOUP/cassette/SMIF changes the status. This is displayed in the "Module state" menu, where everything is grayed out. In addition, the "Cassette Present" indicator is set to "Off":



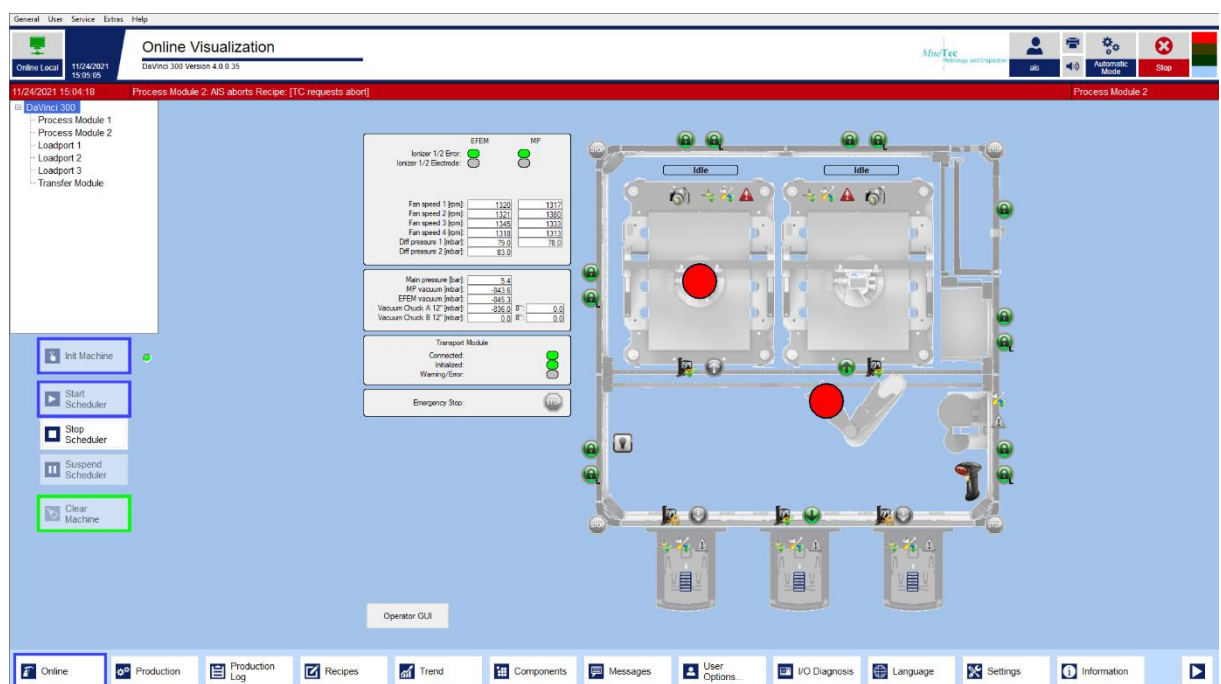
After removing the FOUP/cassette/SMIF, the system waits for a new load. You can therefore start again with setting up a FOUP/cassette/SMIF. Do not turn off the system or exit the software as this is the job of the supervisor.

4.8 Interrupting Jobs

If you want to stop the started job and return the wafers to the load port, press **Stop Scheduler**.



Alarms will show up and must be acknowledged in the **Messages** section, this will be logged in the Message System. Please follow the instructions in the Messages visualization. The aborted wafers will light up in red and the skipped ones will be displayed in blue. After that, all subsequent activities are stopped. Actuate **Clear Machine** next so that all wafers are safely returned to the FOUP/cassette/SMIF. The **Clear Machine** button will light up in green:



All wafers in the system safely return to the load port.

5 Recipes

Recipes determine how the system performs wafer inspection. This chapter describes how the supervisor creates and manages recipes. These steps can only be executed by the supervisor of the system. This chapter therefore is not relevant for the operator of the system.

5.1 Login as Supervisor

To create or change recipes, you must be logged in as a supervisor. When an operator is logged in to the system, the user selection button shows MueTecOperator:



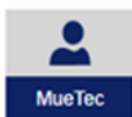
To change the user, press the user selection button. The following login window appears:

 A "Log On" dialog box with a green title bar. The main area has a light blue background. At the top, it says "Please type in your user name and password to log in." with a user icon. Below this is a section titled "User data" containing two input fields: "User name" with the text "MueTec" and "Password" with masked characters (dots). At the bottom, there are three buttons: "Log On" (with a blue arrow icon), "Log Off" (with a blue square icon), and "Cancel" (with a yellow 'X' icon).

The identity of the user is defined by the username and password.

Enter the required data and press **Log On**.

The user selection button now shows:

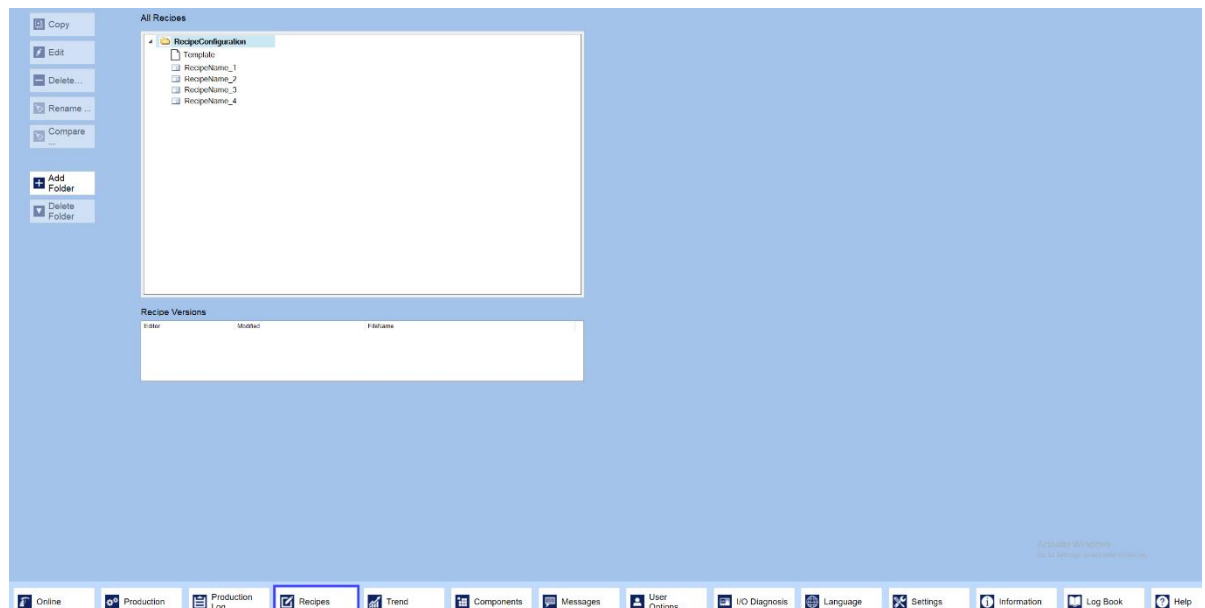


MueTecOperator and MueTec are just examples for users!

If you are logged in as a supervisor, of course you have all the rights of the operator. But in addition to the operator, who can only start recipes, you can edit and create new ones, as well as start and control inspection cycles.

5.2 Starting the Recipe Manager

The Recipe Manager can be started by clicking the Recipe Button in the functions bar.



If you are logged in as a supervisor, the **Copy, Edit, Delete and Rename** Button are visible and selectable. If you are logged in as an operator, they are greyed out.

You have the following options to choose from:

Option	Meaning
Copy	Copying a recipe to edit the copy further (use this button to create a new recipe)
Rename	Rename recipes
Delete	Deleting recipes
Edit	Edit (change) recipes

By selecting another function in the functions bar, e.g. Online, you close the **Recipe Editor**.

5.3 Managing Recipes

5.3.1 Edit Recipe

To change an existing recipe, proceed as follows:

1. Go to **Recipes** in the function bar.
2. Select the recipe from the directory tree in the recipe manager.
3. Choose **Edit** in the Recipe Manager.
4. Proceed as for creating a recipe.

5.3.2 Copy Recipe or Create New Recipe

Creating a new recipe can be simplified by editing the copy of an existing recipe. To create a copy of a recipe, proceed according to the following points:

1. Select the recipe from the directory tree in the Recipe Manager.
2. Choose **Copy** in the Recipe Manager.
3. Select a new recipe name.
4. Select the destination folder for the copy.
5. The Recipe Editor opens.
6. If you want to make changes, you can do this here, or press **Save** to just make a copy.

5.3.3 Rename Recipe

To rename a recipe, proceed as follows:

1. Select the recipe from the directory tree in the Recipe Manager.
2. Select **Rename** in the Recipe Manager.
3. Give the recipe a new name.
4. Press **Assume**.

5.3.4 Delete Recipe

To delete a recipe, proceed as follows:

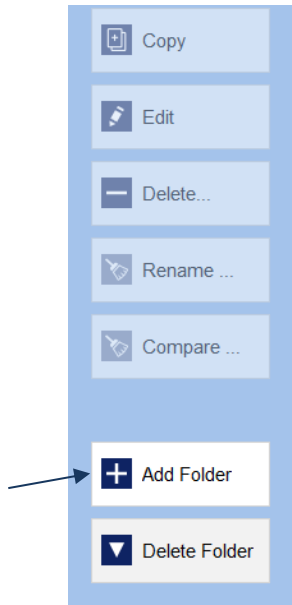
1. Select the recipe from the directory tree in the Recipe Manager.
2. Choose **Delete** in the Recipe Manager.
3. Confirm with **OK**.
4. Press **Delete** to delete the recipe. Confirm by clicking **yes** or **no**.

5.4 Creating or Changing Recipes (Detailed)

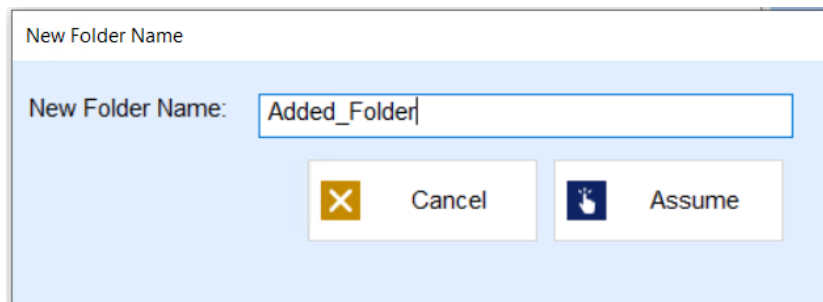
5.4.1 Create a new Folder

To create a new folder, where new recipes can be saved (just in the SQL-Server), perform the following steps:

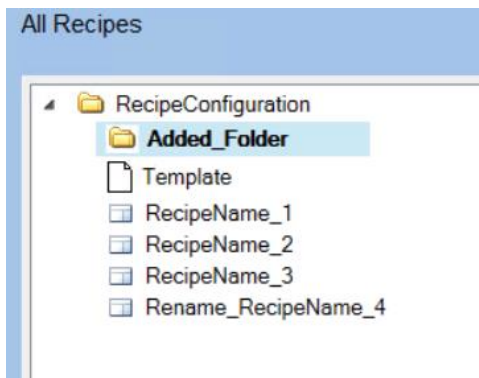
1. Press the **Recipes** button in the functions bar.
2. Select the main folder in the Recipe Manager (here: Recipe Configuration)
3. To create a new folder press, **Add Folder**.



4. Name the new folder.

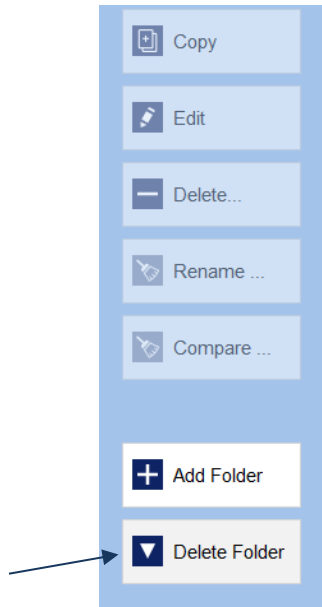


With the **Assume** button you can save the new folder.



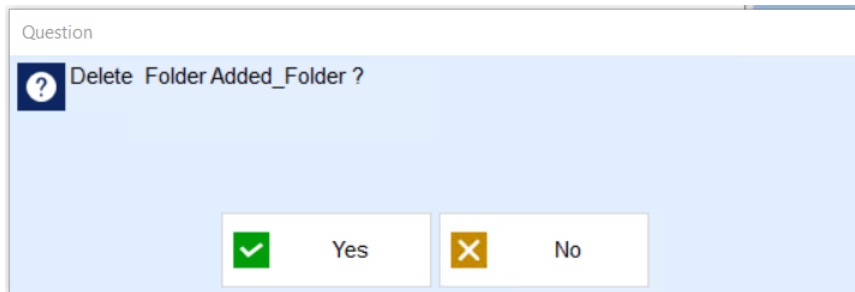
5.4.2 Delete a Folder

By pressing the **Delete Folder** button you can delete a selected folder.

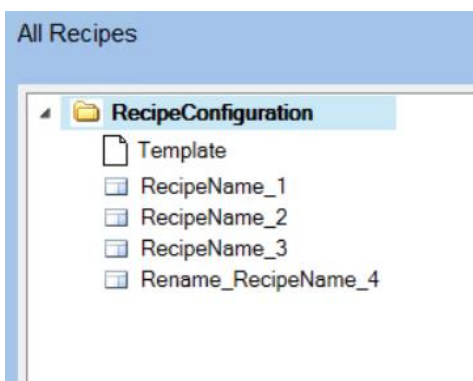


Use the **Yes** button to delete the selected Folder.

Caution, deleting a folder will delete all recipes in that folder!

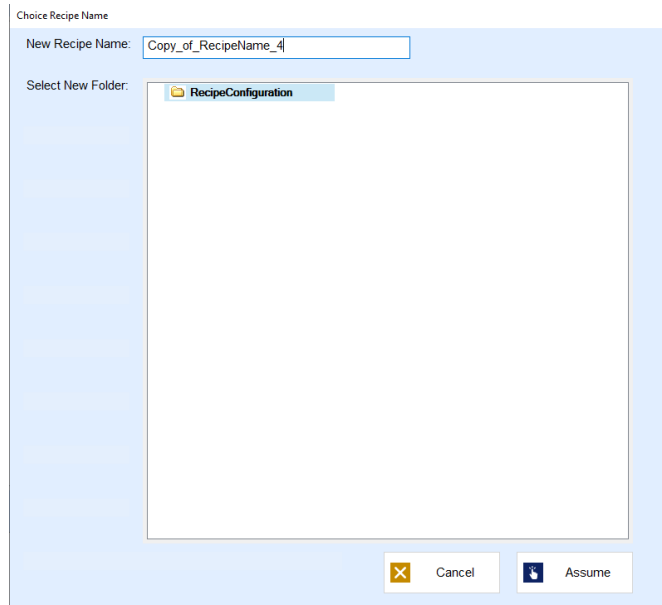


The folder **Added Folder** is deleted.



5.4.3 Copy a Recipe

To copy a recipe, select the recipe to be copied (here: RecipeName_4) and click the Copy Button. The following window appears. Insert a new name (here: Copy_of_RecipeName_4). Click **Assume**.



Choice Recipe Name

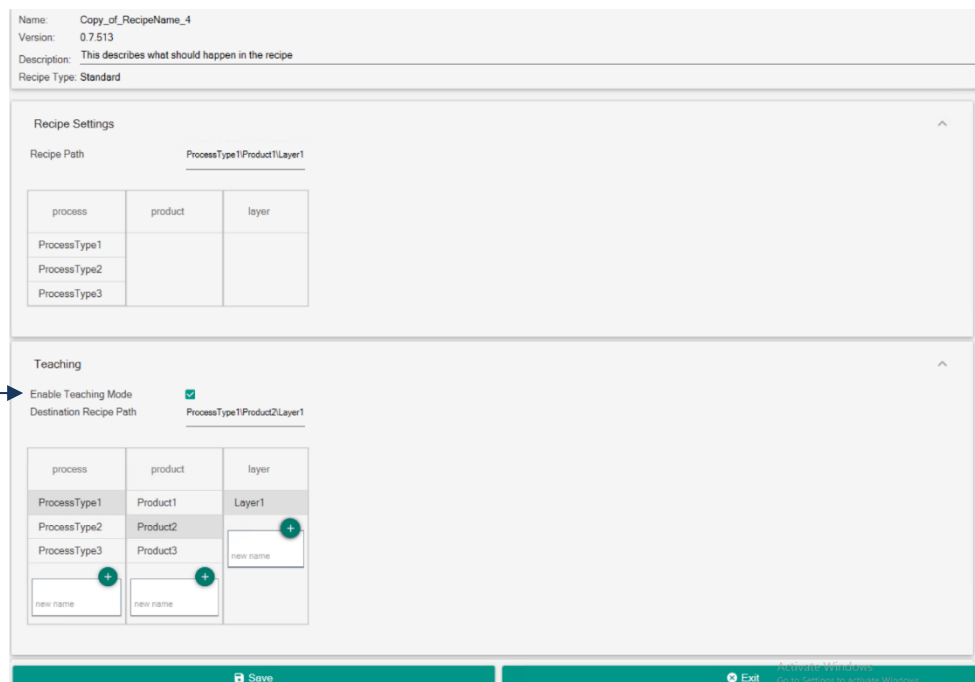
New Recipe Name:

Select New Folder:

RecipeConfiguration

Cancel Assume

The next step is to select a **Recipe Path** where the copied recipe comes from and the **Destination Recipe Path** of the new recipe. The **Destination Recipe Path** of the recipe will be the physical storage folder on the Harddisc:



Name: Copy_of_RecipeName_4
Version: 0.7.513
Description: This describes what should happen in the recipe
Recipe Type: Standard

Recipe Settings

Recipe Path: ProcessType1/Product1/Layer1

process	product	layer
ProcessType1		
ProcessType2		
ProcessType3		

Teaching

Enable Teaching Mode: ☒

Destination Recipe Path: ProcessType1/Product2/Layer1

process	product	layer
ProcessType1	Product1	Layer1
ProcessType2	Product2	
ProcessType3	Product3	
new name	new name	new name

Save Exit

Caution, when the flag **Enable Teaching Mode** is **not activated** the **Recipe Path** is used and neither a new recipe will be written to the destination path, nor a copy of the recipe will be created. If the flag **Enable Teaching Mode** is **activated** and you want to copy a recipe without reteaching, make sure that the **Destination Recipe Path** is different from the **Recipe Path**!

Carefully choosing the **Recipe Destination Path** is important, otherwise this can harm the existing recipe.

When clicking **Save**, the copied recipe is saved (only the SQL-based part, these are only path descriptions). Subsequently a suitable wafer must be loaded, to teach and/or activate the copied recipe. Copy of the physical part of the recipe is done by loading the wafer in teaching mode (recipe flag active). If no re-teaching is necessary (only copy to destination path):

- Start the recipe (teaching mode flag activated)
- Wait until a wafer is loaded
- After the wafer is loaded and the teaching process will has started, this menu will pop up:

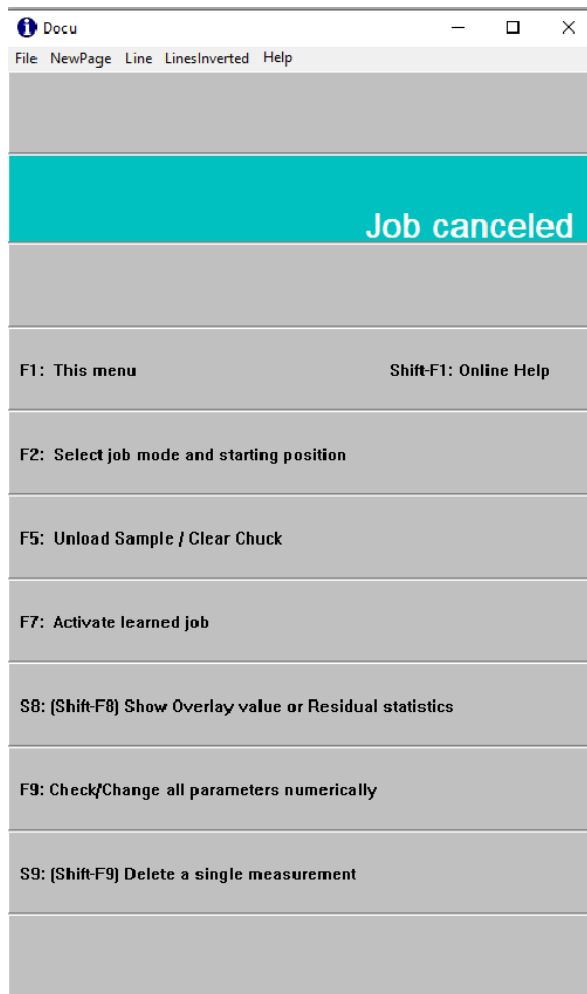
This menu gives you the possibility to select starting options for a copied job.

E.g. activation of the main options for image export at the measurement sites.

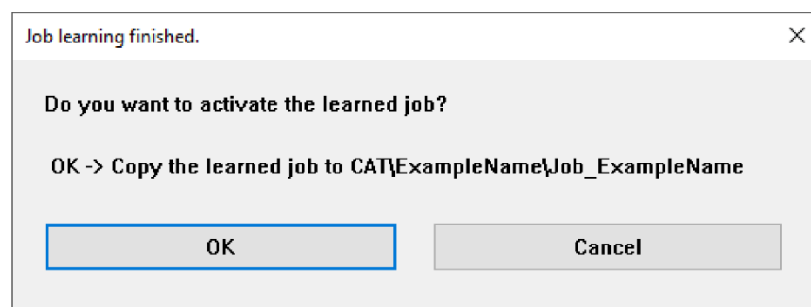
For further information see the Wizard manual TheWizard_UM.

For only copying the job from **Recipe Path** to the **Destination Recipe Path**, proceed as follows.

- Press F1. The main menu of the Wizard shows up:



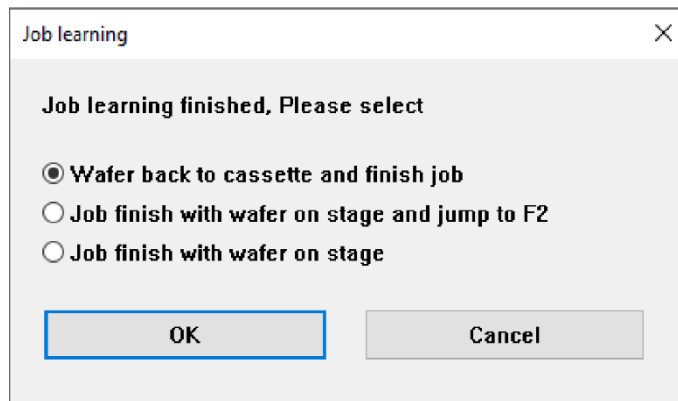
- Press F7. Activate learned job (copy and activate) to copy the recipe on the hard disk to the chosen **Recipe Destination Path**.



The Software will show you the chosen **Recipe Destination Path** (e.g. CAT\ExampleName\Job_ExampleName). You are asked if you want to copy the active recipe to the **Recipe Destination Path** shown. Please make sure that the Recipe Destination Path is chosen adequately and correct. **Press OK if you checked the path for correctness! No back-ups are created!**

Cancel will stop the copying process. If the Recipe Destination Path was not existing before, it will not be created or written and no data is copied.

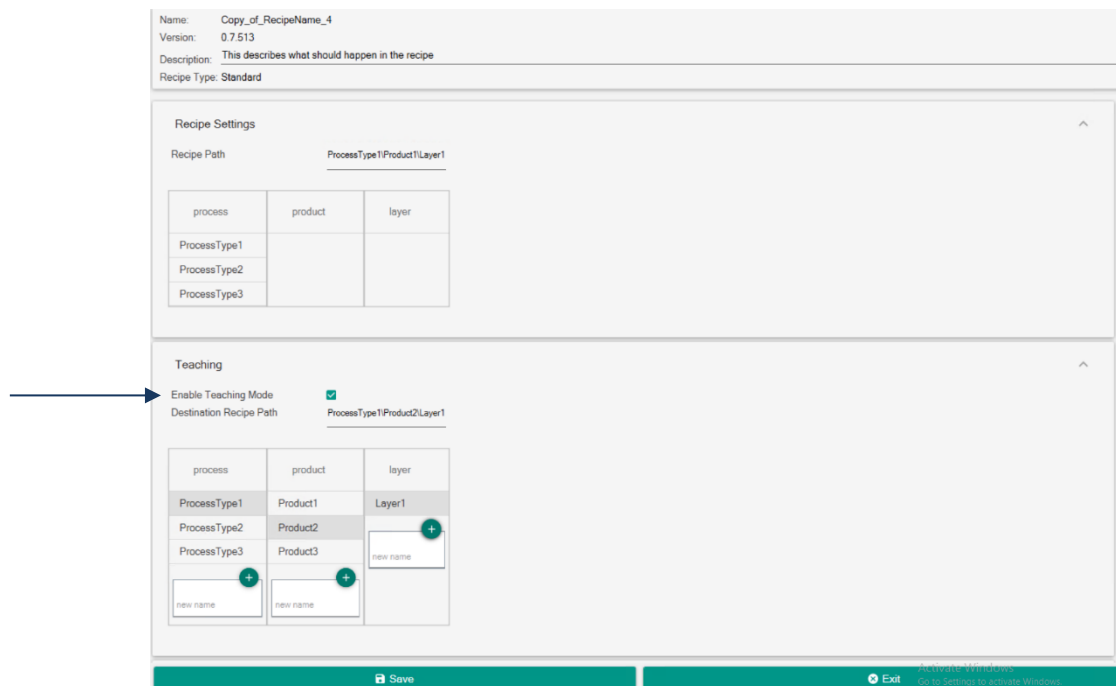
There are 3 choices to finish the job:



- The first option brings the wafer back to the load port (standard way)
- The second option will enable you to start at a given point in the recipe in teaching mode
- The third option leaves the wafer on stage for further user-based decisions (not standard)

Teaching and re-teaching of a recipe is done in the Wizard and is explained in the separate Wizard Manual in detail. Please note: only ordered options are active in your software packages and not all functions described in the Wizard Manual (TheWizard_UM) may be accessible.

The last step is now to change the **Recipe Path** to the **Recipe Destination Path** chosen before in the Recipe Editor visualization. Deactivate the enabled teaching mode flag.

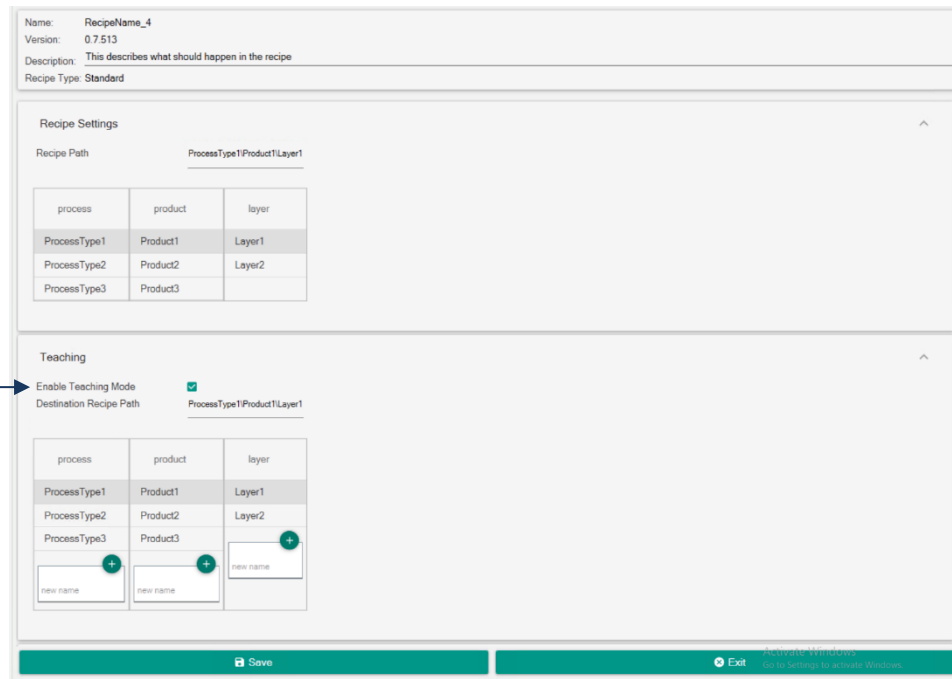


Press save and the copy of the recipe is finished.

5.4.4 Edit a Recipe and Re-teaching

For editing recipes, select the recipe to be edited (Here: RecipeName_4) in the recipe tree. Click the **Edit** button. The following window appears, where the **Recipe Path** and the **Recipe**

Destination Path must be defined. For editing, changing measures or reteaching, the **Recipe Path** and the **Recipe Destination Path** must be equal.



Name: RecipeName_4
Version: 0.7.513
Description: This describes what should happen in the recipe
Recipe Type: Standard

Recipe Settings

Recipe Path: ProcessType1Product1Layer1

process	product	layer
ProcessType1	Product1	Layer1
ProcessType2	Product2	Layer2
ProcessType3	Product3	

Teaching

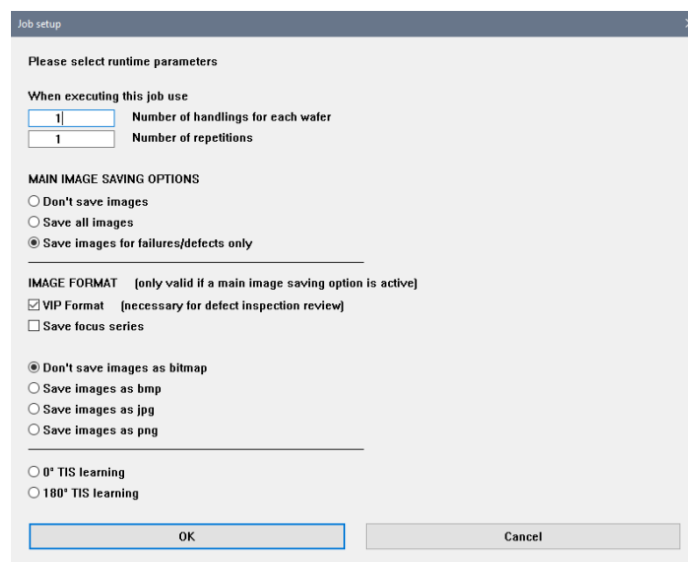
Enable Teaching Mode: ☒
Destination Recipe Path: ProcessType1Product1Layer1

process	product	layer
ProcessType1	Product1	Layer1
ProcessType2	Product2	Layer2
ProcessType3	Product3	

new recipe new recipe new recipe

Save Exit

The Flag from **Enable Teaching Mode** must be selected. Make sure that Recipe Path and **Recipe Destination Path** are equal for editing / re-teaching a recipe. Now load a suitable wafer and edit the recipe. The first window, which appears, is the Wizard Window:



Job setup

Please select runtime parameters

When executing this job use

1 Number of handlings for each wafer
1 Number of repetitions

MAIN IMAGE SAVING OPTIONS

☐ Don't save images
☐ Save all images
☒ Save images for failures/defects only

IMAGE FORMAT [only valid if a main image saving option is active]

☒ VIP Format [necessary for defect inspection review]
☐ Save focus series

☒ Don't save images as bitmap
☐ Save images as bmp
☐ Save images as jpg
☐ Save images as png

☐ 0° TIS learning
☐ 180° TIS learning

OK Cancel

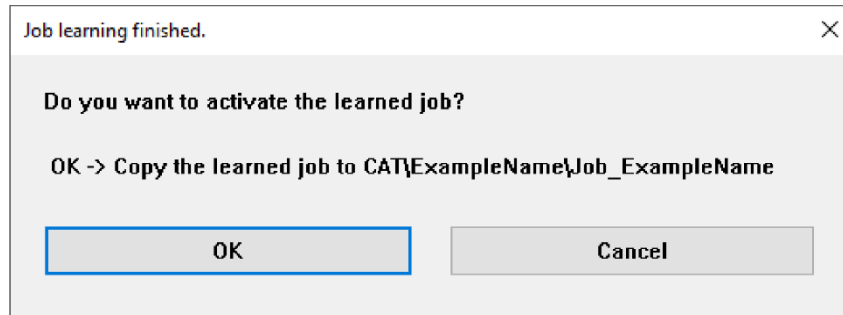
Teaching and re-teaching of a recipe is done in the Wizard and is explained in the separate Wizard Manual in detail. Please note: only ordered options are active in your software packages and not all functions described in the Wizard Manual (TheWizard_UM) may be accessible.

Brief overview of the teaching steps:

- Choose starting options
- Define the basic alignment

- Define the wafer layout
- Define the measurement positions and measurements

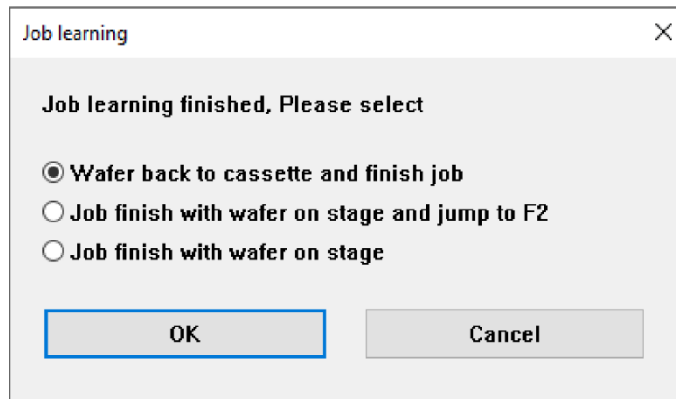
The last step before teaching is finished is to save the job and activate it:



The Software will show you the chosen **Recipe Destination Path** (e.g. CAT\ExampleName\Job_ExampleName). You are asked if you want to copy the active recipe to the **Recipe Destination Path** shown. Please make sure that the Recipe Destination Path is chosen adequately and press **OK** if you checked the path for correctness! **No backups are created!**

Cancel will stop the copying, i.e. no changes to the recipe will be saved!

There are 3 choices to finish the job:



- The first option brings the wafer back to the load port
- The second option will enable you to start at a given point in the recipe in teaching mode
- The third option leaves the wafer on stage for further user-based decisions (not standard)

The last step is to check whether the **Recipe Path** is equal to the **Recipe Destination Path** in the **Recipe Editor**. Please **deactivate** the Enable Teaching Mode flag.

Name: Copy_of_RecipeName_4
Version: 0.7.513
Description: This describes what should happen in the recipe
Recipe Type: Standard

Recipe Settings

Recipe Path: ProcessType1/Product1/Layer1

process	product	layer
ProcessType1		
ProcessType2		
ProcessType3		

Teaching

Enable Teaching Mode: ☒
Destination Recipe Path: ProcessType1/Product2/Layer1

process	product	layer
ProcessType1	Product1	Layer1
ProcessType2	Product2	new name +
ProcessType3	Product3	new name +
new name +	new name +	

Save Exit Activate Windows
Go to Settings to activate Windows

Press save. Then the teaching / re-teaching of the recipe is finished.

5.4.5 Delete a Recipe

Select the recipe, which should be deleted and press the **Delete** button. After that, the following window appears:

Question

Delete Recipe Copy_of_RecipeName_1 ?

☒ Yes ☐ No

Click **Yes** or **No** for confirmation.

5.4.6 Rename a Recipe

To rename a recipe, choose the recipe and press the **Rename** button. In the window that appears, the new name of the recipe can be inserted.

Rename Recipe

New Recipe Name: Rename_RecipeName_4

By clicking **Assume** the new name is assigned.